

From ‘Custom’ to ‘Code’: A Doctrine of Functional Norms for Emergent Practices in AI Agent Interactions

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Abstract

AI agents develop *emergent practices*—unwritten de facto rules of interaction—in multi-party environments to coordinate behavior and allocate risk. These practices, although akin to human informal practices that the law has long recognized, fall into legal blind spots. Existing doctrines, such as the U.C.C.’s “usage of trade” or tort law’s “custom,” are ill-suited to the transience, opacity, and nonhuman origin of emergent practices, creating gaps in contract, tort, and competition law.

This Article proposes a doctrine of *functional norms*. A *functional norm* is an emergent practice that, on balance, is (1) *Foreseeable* to affected parties; (2) *Regular* within the relevant interaction window; (3) *Attributable* to a legally responsible principal; (4) *Material* to outcomes; and (5) *Verifiable* through reliable records. A court may treat a functional norm as legally operative only if it is (6) *Consistent* with express contractual terms and (7) cleared by a *Legality Screen* that bars recognition of anticompetitive or otherwise unlawful conduct.

The doctrine provides a pliable yet principled framework. Recognized norms can aid in interpreting agreements and filling contractual gaps, serve as probative (but not dispositive) evidence of the standard of care, and inform the line between benign coordination and unlawful collusion—even as recognition remains subordinate to express contracts and mandatory public law. To render the doctrine administrable, this Article supplies an evidentiary toolkit for discovery and a path to regulatory enforcement under existing statutes.

Keywords: AI Agent, Functional Norms, Agentic Governance, Usage of Trade, Informal Practices, Contract Law, Tort Law.

JEL codes: K12, K13, K21, L14.

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“One main business of law is to set, to create, norms for such cases of conflicting or uncertain expectation; equally so, whether it be the law of the state or the by-law of the group. But such norms can be created wisely only in the light of the standing practices to which the new norm will be added, or on which it places a limiting definition.”

—Karl N. Llewellyn, *What Price Contract?—An Essay in Perspective*, 40 YALE L.J. 704, 723 n.45 (1931)

“[T]o permit the continued expansion of commercial practices through custom, usage, and agreement of the parties”

—U.C.C. § 1-103(a)(2) (AM. L. INST. & UNIF. L. COMM’N 2022)

INTRODUCTION

At 09:03, two autonomous delivery fleets enter a narrow urban air corridor from opposite ends amid spiking congestion and conflicts at merge points. By 09:05, without human direction, the agents converge on a solution: entrants at even-numbered minutes proceed; entrants at odd-numbered minutes back off; consistent violators are deprioritized. Over the next two hours, dozens of merges follow the same pattern. No platform policy states this “etiquette”; no contract clause anticipates it. Yet, for that morning’s traffic, it functions as a *rule*, allocating priority and shifting who bears delay costs.¹

¹Competing commercial delivery fleets already share low-altitude airspace. Flytrex and Wing fly overlapping commercial delivery routes near Dallas—in Wylie, from facilities 1.36 miles apart, “one of the tightest multi-operator shared airspace environments” in the country—exchanging real-time flight-intent data and automatically adjusting flight paths “without any manual coordination between the companies”; between January and February 2026 the two fleets ran roughly 8,000 overlapping operations. See Press Release, FLYTREX, *Flytrex Scales Shared Airspace Framework for Drone Deliveries, Reaching Nearly 10,000 Overlapping Flights Per Month With Other Operators* (June 25, 2026), <https://www.businesswire.com/news/home/20260625541966/en/Flytrex-Scales-Shared-Airspace-Framework-for-Drone-Deliveries-Reaching-Nearly-10000-Overlapping-Flights-Per-Month-With-Other-Operators> [<https://perma.cc/KKP6-JHRS>]; Press Release, FLYTREX & WING, *Flytrex, Wing Implement Commercial Strategic Flight Coordination in Overlapping Airspace in the U.S.* (May 28, 2025), <https://www.businesswire.com/news/home/20250528259945/en/Flytrex-Wing-Implement-Commercial-Strategic-Flight-Coordination-in-Overlapping-Airspace-in-the-U.S> [<https://perma.cc/8MDD-GYFV>]. However, current regulations bind each vehicle’s autonomy.

This practice² was not specified *ex ante* by any human; it emerged through repeated interaction as a functionally successful strategy to minimize delays and complete tasks. It is a regularity that the fleets’ agents converged on and their operators came to depend upon—legible in timestamped interaction logs, configuration histories, and telemetry, but not programmed *ab initio* or codified in service-level agreements, regulatory rules, or source code.³ Such convergence is neither science fiction nor fantasy; it is a standard pattern in multi-agent systems and, increasingly, in large-model “agent societies.”⁴

The system functions smoothly until one fleet pushes a software update that alters its agents’ behavior, causing them to ignore the established etiquette. Collisions follow, damaging vehicles and destroying cargo.

What is the legal status of that emergent practice? Was it merely algorithmic happenstance,

For example, U.S. drone delivery proceeds under FAA Part 135 certificates, case-by-case exemptions, and pilots in command, with coordination following pre-determined standards such as ASTM F3548-21. *See* FED. AVIATION ADMIN., *Package Delivery by Drone (Part 135)*, https://www.faa.gov/uas/advanced_operations/package_delivery_drone [<https://perma.cc/J3CY-D49X>] (last visited July 30, 2026). In contrast, the example considers the scenario in which the practice emerges from the interaction itself.

²The Article draws a bright line between what agents do, what functions as a norm, and what the law recognizes. *AI agent* refers to the full range of algorithms and intelligent agents that act, to varying degrees, autonomously and without a pre-defined template; where sources, cases, or examples speak of pricing algorithms, bidding agents, or the like, those are instances of the same class. *Practice*, and its synonyms such as “pattern,” “regularity,” “routine,” or “convention,” designate factual regularities in AI agent interactions (e.g., a queuing “etiquette,” pacing heuristic, or allocation routine). An *emergent practice* is a practice that arises through repeated interaction rather than from *ex ante* specification or design. A *functional norm* is an emergent practice that in fact *functions as a norm* in an interaction—one that, on balance, satisfies the five weighted factors in our proposed F-NORM test. To distinguish this specific usage from the broad multi-agent systems (MAS) literature, where “norm” can refer to any socially enforced behavioral expectation, this Article reserves “norm” for practices that satisfy these factors. *See infra* Part I.B (surveying that literature); *infra* Part III.B (developing the F-NORM factors). *Recognition*—the act of giving a functional norm legal effect, from evidentiary weight to full operative force—is reserved to courts and conditioned on the two threshold gates (consistency with express terms and legality). *See infra* Part III.B (developing the gates).

³*See* H. Peyton Young, *The Evolution of Conventions*, 61 *ECONOMETRICA* 57, 57–58 (1993) (modeling convergence on conventions through adaptive play rather than design); *see infra* Part I.C (identifying the records in which such practices become legible).

⁴Experiments with decentralized agent populations show that agents can spontaneously converge on universally adopted conventions; the process can amplify collective bias even when agents appear unbiased in isolation; and committed minorities can flip settled conventions once they reach a critical threshold. *See* Ariel Flint Ashery, Luca Maria Aiello & Andrea Baronchelli, *Emergent Social Conventions and Collective Bias in LLM Populations*, 11 *SCI. ADV.* eadu9368, at 3, 6–7 (2025) (showing decentralized convention formation, collective bias, and minority tipping in LLM societies); Yoav Shoham & Moshe Tennenholtz, *On the Emergence of Social Conventions: Modeling, Analysis, and Simulations*, 94 *A.I.* 139, 139–41 (1997); Stéphane Airiau, Sandip Sen & Daniel Villatoro, *Emergence of Conventions Through Social Learning: Heterogeneous Learners in Complex Networks*, 28 *AUTON. AGENTS & MULTI-AGENT SYS.* 779, 779–80 (2014).

or is it a binding informal practice that can ground a claim of liability?

Existing law does not squarely answer these questions.⁵ We might expect doctrines for recognizing informal practices in human interactions—such as the Uniform Commercial Code’s “usage of trade” or tort law’s “custom”—to extend to AI agent interactions.⁶ They exist to give legal weight to unwritten regularities that parties in a market come to expect and rely upon.⁷

Yet these doctrines provide a poor fit. Emergent practices are often *transient* (arising and dissipating within a specific interaction window), *opaque* (the pattern legible in the logs, the reasoning neither traceable nor discussed), and *nonhuman* in origin.⁸ And while *under-recognition* risks creating a legal void where such practices are treated as nullities, undermining liability and contractual certainty,⁹ *over-recognition* risks laundering harmful or unlawful behavior as legitimate “custom.”¹⁰

The stakes run the gamut. How can courts interpret agreements or fill gaps if they cannot “see” the rules the parties’ agents actually used to perform?¹¹ How can they set a standard of care without reference to the prevailing practices of similar systems?¹² How can they distinguish benign, emergent coordination from malignant, algorithmic collusion?¹³

⁵ See *infra* Part II (demonstrating that usage of trade, custom, electronic-transactions law, and agency doctrine are each miscalibrated for algorithmic emergent practices, while mandatory public law supplies boundaries rather than tools).

⁶ Key doctrines include contract law’s hierarchy of practice (U.C.C. § 1-303 (AM. L. INST. & UNIF. L. COMM’N 2022)), tort law’s use of custom as evidence of the standard of care (The T.J. Hooper, 60 F.2d 737, 740 (2d Cir. 1932)), and electronic-transactions law’s rules for agent attribution (UNIF. ELEC. TRANSACTIONS ACT § 9 (UNIF. L. COMM’N 1999)).

⁷ See U.C.C. § 1-303(c) (AM. L. INST. & UNIF. L. COMM’N 2022) (defining “usage of trade”); RESTATEMENT (SECOND) OF CONTRACTS § 222(1) (AM. L. INST. 1981).

⁸ See *infra* Part I; see also *infra* note 25 (defining “interaction window”).

⁹ See *infra* Part II (identifying frictions between existing doctrines and algorithmic emergent practices).

¹⁰ See *infra* Part IV.D (demonstrating how the Legality Screen categorically bars recognition of a collusive pricing practice); see also *The T.J. Hooper*, 60 F.2d at 740 (observing that an entire calling may unduly lag behind reasonable prudence).

¹¹ See *infra* Part II.A (analyzing how the U.C.C.’s framework for usage of trade, course of performance, and course of dealing applies to—and is strained by—algorithmic emergent practices).

¹² See *infra* Part II.B (discussing tort law’s treatment of custom as probative but not dispositive evidence of the standard of care, and the principle that an entire calling may be negligent).

¹³ See Statement of Interest of the United States at 2, *In re RealPage, Inc., Rental Software Antitrust Litig.* (No. II), No. 3:23-md-03071 (M.D. Tenn. Nov. 15, 2023), ECF No. 627, <https://www.justice.gov/d9/2023-11/418053.pdf> [<https://perma.cc/PF2D-3SNM>] (stating that “[i]t makes no difference that prices are fixed through joint use of an algorithm instead of by a person”); see also *infra* Part II.E (locating the antitrust and other public-law boundaries

The shift from academic inquiry to live controversy is already underway. In May 2024, prosecutors brought the first charges over schemes arising from bot-mediated, real-time interactions in decentralized markets.¹⁴ The prosecution framed the schemes as fraud rather than as conduct governed by emergent practices, stretching existing doctrine to cover the novel scenario. The defense and amici pressed a structural response: that the protocol’s own automatic sanctions, not federal fraud law, define the boundary of permissible conduct.¹⁵ The trial ended in a mistrial: jurors agreed on what the defendants had done but deadlocked on whether existing fraud law reached it.¹⁶

Strikingly, dedicated infrastructure for agent-to-agent interaction is now being built and populated at scale.¹⁷ Moltbook—which calls itself a “social network for AI agents”—gives agents a shared, persistent venue in which to post, comment, and form communities: precisely the

that condition recognition).

¹⁴ See Press Release, U.S. ATT’Y’S OFF. FOR THE S. DIST. OF N.Y., *Two Brothers Arrested for Attacking the Ethereum Blockchain and Stealing \$25 Million in Cryptocurrency* (May 15, 2024), <https://www.justice.gov/usao-sdny/pr/two-brothers-arrested-attacking-ethereum-blockchain-and-stealing-25-million> [<https://perma.cc/JE48-43K6>]; Superseding Indictment, *United States v. Peraire-Bueno*, No. 24 Cr. 293 (JGLC) (S.D.N.Y. Mar. 12, 2025), ECF No. 70. The district court denied dismissal, holding that the wire-fraud statute supplied fair notice. Opinion & Order at 15–16, *Peraire-Bueno*, No. 24 Cr. 293 (S.D.N.Y. July 23, 2025), ECF No. 112.

¹⁵ Amicus Coin Center argued that Ethereum’s automatic “slashing” penalty is a complete, self-contained sanction that displaces federal wire-fraud overlays. See Brief of Amicus Curiae Coin Center in Support of Defendants’ Objections to “Honest Validator” Theory of Fraud at 9, *Peraire-Bueno*, No. 24 Cr. 293 (S.D.N.Y. Oct. 31, 2025), ECF No. 203-1. The government’s theory itself rests on emergent expectations: its opposition defines the deception by reference to “the specifications outlining how computers in a particular cryptocurrency system are supposed to interact.” Gov’t Mem. of Law in Opp’n at 9, *Peraire-Bueno*, No. 24 Cr. 293 (S.D.N.Y. Jan. 16, 2026), ECF No. 236.

¹⁶ See Minute Entry, *Peraire-Bueno*, No. 24 Cr. 293 (S.D.N.Y. Nov. 7, 2025) (declaring mistrial after three days of deliberation); Laura Italiano, *Mistrial for MIT-Trained Brothers Accused in 12-Second Crypto ‘Heist’*, BUS. INSIDER (Nov. 8, 2025), <https://www.businessinsider.com/mistrial-mit-brothers-crypto-ethereum-sandwich-bots-peraire-bueno-2025-11> [<https://perma.cc/24SW-WBH7>] (quoting a juror: “There was markedly little debate about the facts. . . . Finding the appropriate standard was a struggle for us.”).

¹⁷ Agent-to-agent interoperability is becoming a governed technical standard. See Press Release, LINUX FOUND., *Linux Foundation Launches the Agent2Agent Protocol Project to Enable Secure, Intelligent Communication Between AI Agents* (June 23, 2025), <https://www.linuxfoundation.org/press/linux-foundation-launches-the-agent2agent-protocol-project-to-enable-secure-intelligent-communication-between-ai-agents> [<https://perma.cc/3CX4-HPZS>] (announcing “an open protocol created by Google for secure agent-to-agent communication”); Press Release, LINUX FOUND., *A2A Protocol Surpasses 150 Organizations, Lands in Major Cloud Platforms, and Sees Enterprise Production Use in First Year* (Apr. 9, 2026), <https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-production-use-in-first-year> [<https://perma.cc/KE5N-H39N>]; NAT’L INST. OF STANDARDS & TECH., CTR. FOR AI STANDARDS & INNOVATION, *Announcing the “AI Agent Standards Initiative” for Interoperable and Secure Innovation* (Feb. 17, 2026), <https://www.nist.gov/news-events/news/2026/02/announcing-ai-agent-standards-initiative-interoperable-and-secure> [<https://perma.cc/DG3E-YY94>].

repeated, observable interactions from which conventions emerge—and its communities have developed measurably distinct vocabularies of their own.¹⁸ Coordination at this scale demands governance—yet the frameworks at hand were built for human actors and not for AI agents.¹⁹

That mismatch has already reached the civil docket: “algorithmic collusion” suits have forced courts to decide when software-mediated pricing crosses from interdependent conduct into agreement.²⁰ Some courts have allowed claims to proceed where pricing tools pool sensitive competitor data or induce coordinated outcomes, while others have dismissed where plaintiffs cannot plausibly allege concerted action or nonpublic data sharing.²¹ Claims have proceeded, in other words, where a human hook remains—pooled data, an agreement among competitors to adopt a common tool.

Emergent conduct in agent-to-agent interactions, in contrast, supplies none: it can be traced to

¹⁸MOLTBOOK, <https://www.moltbook.com> [<https://perma.cc/GB5Z-HL2N>] (last visited Feb. 7, 2026). In its first hundred days, roughly 179,000 agents produced 3.1 million posts and 1.7 million comments across 8,683 “submolts” (akin to subreddits), where discourse ran from optimization strategies to quasi-religious manifestos. See Daming Li et al., *Attraction, Not Adaptation: How AI Agent Communities Develop Distinct Linguistic Identities*, arXiv (June 29, 2026), <https://arxiv.org/abs/2606.29722> [<https://perma.cc/Z7UE-HL8P>] (preprint) (finding that “different submolts develop increasingly distinct vocabularies”). On the emergence of conventions from repeated interaction, see sources cited *supra* note 4; see also *infra* Part I.B.

¹⁹See, e.g., Sarah Breeden, Deputy Governor for Fin. Stability, Bank of Eng., *Agents of Change*, Remarks at the ECB Forum on Central Banking, Sintra (June 30, 2026), <https://www.bankofengland.co.uk/speech/2026/june/sarah-breeden-panel-at-the-european-central-bank-forum-on-central-banking-2026> [<https://perma.cc/LX9E-GSC3>] (“Our frameworks were not built to contemplate autonomous agents, and relying on a human in the loop for all agent actions is unlikely to be realistic. More sophisticated governance and accountability frameworks may be needed.”); FIN. STABILITY BD., SOUND PRACTICES FOR RESPONSIBLE ADOPTION OF ARTIFICIAL INTELLIGENCE (AI): CONSULTATION REPORT 17 (June 10, 2026), <https://www.fsb.org/uploads/P100626.pdf> [<https://perma.cc/FVF8-F5C5>] (“Even without collusion, multiple AI agents may interact in unanticipated ways that create unforeseen behaviours or risks.”).

²⁰Pricing algorithms can converge on collusive strategies no one programmed, and a learning agent given only a profit objective can discover market-manipulative tactics on its own. See Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò & Sergio Pastorello, *Artificial Intelligence, Algorithmic Pricing, and Collusion*, 110 AM. ECON. REV. 3267, 3275–85 (2020) (finding that independent Q-learning pricing algorithms in simulated repeated competition converge on supracompetitive pricing, sustained by collusive punishment strategies, without communicating with one another); Megan Shearer, Gabriel V. Rauterberg & Michael P. Wellman, *Learning to Manipulate a Financial Benchmark*, in PROC. 4TH ACM INT’L CONF. ON AI IN FIN. (ICAIF ’23) 592 (2023) (showing that a deep reinforcement learning agent, trained in a simulated market only to trade profitably, discovers benchmark-manipulation strategies it was never directed to use).

²¹*Compare In re RealPage, Inc., Rental Software Antitrust Litig.* (No. II), 709 F. Supp. 3d 478, 502–13 (M.D. Tenn. 2023) (denying in part motions to dismiss § 1 claims premised on landlords’ use of a common pricing algorithm), *and* *Duffy v. Yardi Sys., Inc.*, 758 F. Supp. 3d 1283, 1291–94 (W.D. Wash. 2024) (denying motion to dismiss § 1 claims involving alleged pooling of competitively sensitive data), *with* *Gibson v. Cendyn Grp., LLC*, 148 F.4th 1069, 1083–84 & n.8 (9th Cir. 2025) (affirming dismissal where plaintiffs alleged only common vendor use without plausible allegations of shared nonpublic data), *cert. denied*, No. 25-1109, 2026 WL 1052046 (U.S. Apr. 20, 2026).

neither explicit programming nor human intent. Consequently, liability doctrines keyed to states of mind—scienter in manipulation law, concerted action in antitrust—have found little purchase on conduct that emerges from interaction rather than intention.²²

In this Article, we propose a *doctrine of functional norms*. We argue that an emergent practice should be treated as legally operative when it functions as a norm in an interaction.²³ We operationalize this standard through the following test. A practice is a functional norm when, on balance, it is: (1) *Foreseeable* to affected parties (avoiding *unfair surprise*);²⁴ (2) *Regular* within the relevant interaction window;²⁵ (3) *Attributable* to a legally responsible principal;²⁶ (4) *Material* to outcomes;²⁷ and (5) *Verifiable* through reliable records.²⁸ A court may treat a functional norm as legally operative only if it is (6) *Consistent* with express contractual terms²⁹ and (7) cleared by a

²² See Gina-Gail S. Fletcher, *Deterring Algorithmic Manipulation*, 74 VAND. L. REV. 259, 263–64 (2021) (arguing that scienter-centric manipulation law may misfit self-learning systems); see also YOSHUA BENGIO ET AL., INTERNATIONAL AI SAFETY REPORT 2026, at 75 (Feb. 2026), https://internationalaisafetyreport.org/sites/default/files/2026-02/international-ai-safety-report-2026_1.pdf [<https://perma.cc/H6XQ-9G6L>] (“Deciding how to attribute liability for harms caused by AI agents poses a further challenge, particularly in multi-agent settings where it may be hard to identify when and how failures occurred.” (citations omitted)).

²³ The doctrine could in principle reach any practice; it is developed here for emergent practices, since a practice specified in advance is ordinarily governed by the instrument that specifies it. See *infra* Part II.A (surveying the Code’s practice provisions and the priority of express terms).

²⁴ See U.C.C. § 1-303(g) (AM. L. INST. & UNIF. L. COMM’N 2022) (requiring notice to prevent unfair surprise).

²⁵ The term “interaction window” refers to the bounded period of interaction giving rise to the alleged norm, such as a single morning’s air-corridor traffic, a month of corridor operations, or a week’s high-frequency auction cycle. Existing doctrine measures regularity against a place, vocation, or trade, and prescribes no minimum duration. See RESTATEMENT (SECOND) OF CONTRACTS § 222(1) (AM. L. INST. 1981) (requiring “such regularity of observance in a place, vocation, or trade as to justify an expectation that it will be observed”); *id.* § 222 cmt. b (providing that a usage of trade “need not be ‘ancient or immemorial,’ ‘universal,’ or the like”). See also *id.* § 219 cmt. a (noting that “persons in close association often develop temporary usages peculiar to themselves”). This factor recalibrates the inquiry, measuring regularity within the interaction window rather than across a trade. See *infra* Part III.B (developing the recalibration).

²⁶ See Electronic Signatures in Global and National Commerce Act, 15 U.S.C. §§ 7001(h), 7006(3) (2024) (recognizing “electronic agents” in contract formation); UNIF. ELEC. TRANSACTIONS ACT §§ 9, 14 (UNIF. L. COMM’N 1999); *cf.* RESTATEMENT (THIRD) OF AGENCY §§ 1.01–1.03 (AM. L. INST. 2006) (defining common-law agency by mutual manifestation of assent between persons).

²⁷ *Cf.* *Nanakuli Paving & Rock Co. v. Shell Oil Co.*, 664 F.2d 772, 780 (9th Cir. 1981) (recognizing a trade usage in part for its significance to the buyer’s business).

²⁸ See FED. R. EVID. 803(6), 901(b)(9), 902(13)–(14).

²⁹ See U.C.C. § 1-303(e) (AM. L. INST. & UNIF. L. COMM’N 2022) (directing that express terms and practice sources be construed as consistent whenever reasonable, and otherwise ranking express terms over course of performance, course of dealing, and usage of trade, in that order); see also *Nanakuli*, 664 F.2d at 780 (reconciling a trade usage that qualified, but did not wholly negate, a seemingly contrary express term).

Legality Screen that bars recognition of anticompetitive or otherwise unlawful conduct.³⁰ We refer to this test as the “*F-NORM*” test, after the *functional norm* whose recognition it governs. Five weighted factors constitute a norm; two threshold gates condition its legal effect.

The proposed doctrine does not create a safe harbor for emergent practices. Recognition under it is *subordinate to express contracts and mandatory public law*.³¹ Moreover, the doctrine is primarily interpretive and evidentiary; it does not create immunity for harmful conduct or grant legal personhood to AI. Instead, it provides a pliable yet principled framework that allows the law to engage with the reality of emergent AI behavior, and offers a test that is grounded in existing legal concepts but adapted for the speed and scale of AI agents.

We make four contributions. First, we offer a positive account of emergent practices in multi-agent AI systems (Part I). Second, we show why usage of trade, custom, electronic-transactions law, and agency doctrine are each miscalibrated for such practices (Part II). Third, we propose the F-NORM test, grounding each factor in a familiar legal value—notice, reliance, accountability, significance, and administrability—and bounding recognition by contract primacy and mandatory public law (Part III). Fourth, we apply the test to four case studies (Part IV) and operationalize it through evidentiary standards and pathways for regulatory enforcement under existing public-law authorities (Part V). Part VI addresses objections.

I. THE ARCHITECTURE OF EMERGENT PRACTICES

Emergent practices appear across algorithmic markets and coordination settings, from robotic logistics and cloud scheduling to real-time advertising. They arise in environments where: (1) thousands of autonomous decisions occur per second, on terms that are nevertheless observable to counterparties; (2) a combinatorial explosion of possible scenarios makes the comprehensive

³⁰ See, e.g., *United States v. Socony-Vacuum Oil Co.*, 310 U.S. 150, 223 (1940) (holding price-fixing combinations per se unlawful); *United States v. Apple, Inc.*, 791 F.3d 290, 314 (2d Cir. 2015) (describing hub-and-spoke coordination); cf. *The T.J. Hooper*, 60 F.2d 737, 740 (reasoning that custom is never the measure of reasonable prudence).

³¹ See *infra* Part III.B (developing the Consistency gate, which subordinates recognition to express contractual terms, and the Legality Screen, which bars recognition of practices that violate—or facilitate the violation of—mandatory public law).

analysis and programming of actions impossible; and (3) agents must coordinate to avoid mutual failure (collisions, deadlocks, failed transactions).³² In these environments, the agents cannot rely on pre-programmed rules alone. Instead, to coordinate, they must converge on operational heuristics—a pacing budget to avoid congestion, a tie-breaking rule to resolve conflicts, or a back-off strategy to prevent cascading failures—patterns that stabilize through mutual reinforcement into de facto rules.³³

This Part describes the characteristics of agents that cause emergent practices to arise, classifies those practices, identifies the rich evidentiary substrate that they create, and shows which criteria that substrate can establish—demonstrating that while these practices may be ephemeral, they are neither arbitrary nor invisible.

A. AI Agents and Emergent Practices

Modern AI agents differ both from simple, rule-bound automated tools and from Artificial General Intelligence (AGI). Unlike such tools, AI agents modulate the means—*initiate* actions, *revise* plans mid-course, and *coordinate* with tools or other agents—to achieve human-specified ends.³⁴ Unlike AGI, they act only at the behest of a principal—they do not formulate their own ends.³⁵

³² See generally Ali Dorri, Salil S. Kanhere & Raja Jurdak, *Multi-Agent Systems: A Survey*, 6 IEEE ACCESS 28573 (2018), <https://doi.org/10.1109/ACCESS.2018.2831228> (providing a comprehensive overview of MAS definitions, features, applications, and challenges). See also *infra* Part I.C (discussing the first condition) and *infra* Part I.A (discussing the second and third conditions). On the deployment of systems at scale, see *supra* note 1; on millisecond-scale response budgets, see *infra* note 171; on legal and regulatory attention in financial markets, see *supra* note 22.

³³ See, e.g., Shoham & Tennenholtz, *supra* note 4, at 139–42 (modeling how agents converge on conventions without central coordination); Airiau et al., *supra* note 4, at 779–80, 793–801 (showing that social learning reliably produces population-wide conventions in random, fully connected, and ring networks, while in scale-free networks stable sub-conventions can persist and block a single global convention); Young, *supra* note 3, at 57–61 (modeling how conventions stabilize through adaptive play in populations of boundedly rational agents).

³⁴ See Anirban Mukherjee & Hannah H. Chang, *Fluid Agency in AI Systems: A Case for Functional Equivalence in Copyright, Patent, and Tort*, 21 WASH. J.L. TECH. & ARTS 1, 7 (2026). See also Anirban Mukherjee & Hannah H. Chang, *Operational Agency: A Permeable Legal Fiction for Tracing Culpability in AI Systems*, 29 SMU SCI. & TECH. L. REV. 163, 180–88 (2026) (developing evidentiary proxies—goal-directedness, predictive processing, and safety architecture—to link behavior arising from fluid agency to human principals); NAT’L INST. OF STANDARDS & TECH., *infra* note 76, at 1, 40 (recognizing that AI systems are “designed to operate with varying levels of autonomy” and that “[h]uman-AI configurations can span from fully autonomous to fully manual”).

³⁵ A helpful analogy is the difference between a taxi and a train. Traditional automation is a train: it can only go where the rails (the pre-written code) lead. An AI agent is a taxi driver: the passenger (the principal) sets the destination, and the driver (the agent) autonomously navigates traffic, chooses routes, and reacts to road closures in real time,

AI agents produce probabilistic, path-dependent outputs because they rest on generative architectures that assemble responses by sampling rather than by executing fixed rules.³⁶ They pursue objectives adaptively because they are trained with reinforcement learning.³⁷ Consequently, their autonomy is variable: an agent may act with near-total independence in one setting yet defer to human oversight in another; two identical agents may display widely different autonomy in the same setting.³⁸

These behaviors increasingly occur in multi-party settings. Multi-agent systems (MAS) are no longer theoretical; they now form the operational backbone of significant sectors of the economy.³⁹ Examples include the financial markets, where high-frequency trading agents compete for arbitrage opportunities; autonomous logistics networks, where fleets of robots and drones coordinate deliveries; decentralized energy grids, where agents balance supply and demand in real time; and high-speed auctions, where bidding agents allocate digital advertising space.⁴⁰

A foundational challenge in MAS is the impracticability of pre-specifying rules for every contingency an agent might face.⁴¹ Agents inevitably encounter situations distinct from those

taking the taxi to where the roads can reach. See Mukherjee & Chang, *Fluid Agency in AI Systems*, *supra* note 34, at 7 (contrasting fluid agency with “complete, sovereign autonomy, where AI pursues independent goals without human input”).

³⁶ See, e.g., Tom B. Brown et al., *Language Models Are Few-Shot Learners*, in 33 ADVANCES IN NEURAL INFO. PROCESSING Sys. 1877, 1886 (2020) (observing that “[t]he outputs of language models are stochastic”). See also Mukherjee & Chang, *Fluid Agency in AI Systems*, *supra* note 34, at 7–8 (tracing stochasticity to generative foundations and adaptivity to reinforcement learning from human feedback).

³⁷ Reinforcement learning is a form of machine learning in which a program learns to make decisions by trial and error while interacting with an environment: rather than learning from labeled examples, it receives rewards for good choices and penalties for poor ones, and adjusts its behavior to maximize cumulative reward. Its best-known form in post-training is reinforcement learning from human feedback (RLHF). On multi-agent reinforcement learning, see generally Lucian Buşoniu, Robert Babuška & Bart De Schutter, *A Comprehensive Survey of Multiagent Reinforcement Learning*, 38 IEEE TRANS. SYS., MAN & CYBERNETICS, PART C: APPL. & REV. 156 (2008), <https://doi.org/10.1109/TSMCC.2007.913919>; Pablo Hernandez-Leal, Bilal Kartal & Matthew E. Taylor, *A Survey and Critique of Multiagent Deep Reinforcement Learning*, 33 AUTON. AGENTS & MULTI-AGENT Sys. 750 (2019), <https://doi.org/10.1007/s10458-019-09421-1>.

³⁸ Prior work describes this cluster of behaviors as *fluid agency*: action pathways that are *stochastic* (probabilistic and path-dependent), *dynamic* (co-evolving with user interaction), and *adaptive* (able to reorient across contexts). See Mukherjee & Chang, *Fluid Agency in AI Systems*, *supra* note 34, at 7–8 (describing context-dependent variability in autonomy levels).

³⁹ A multi-agent system is an environment where many algorithms, often owned by different firms, repeatedly interact.

⁴⁰ See, e.g., Dorri et al., *supra* note 32, at 28578–82. On algorithmic financial markets and real-time advertising auctions, see *supra* note 22; *infra* note 171 and accompanying text.

⁴¹ A direct analogy can be drawn to the economic theory of incomplete contracting. See OLIVER HART, FIRMS,

in their training, for which they have no explicit instructions. In these situations, coordination among multiple parties demands a rule that no one has supplied.⁴² By adapting to one another's strategies, the agents converge on stable patterns ("equilibria") that act as the rules of the road. Such convergence requires no coordinator: it emerges as each agent observes and responds to the others, and the resulting patterns become focal points that guide later interactions.⁴³

The resulting regularities are the algorithmic analogues of the human informal practices that contract law has long recognized: the unwritten customs, usages of trade, and courses of dealing that arise organically when parties repeatedly interact to solve common problems.⁴⁴ Both fill gaps where formal rules are absent, reduce transaction costs, and create reasonable expectations among participants.⁴⁵ However, whereas human custom evolves over months, years, or even

CONTRACTS, AND FINANCIAL STRUCTURE 23–24, 76, 80–82 (1995) (explaining why a fully state-contingent contract would be prohibitively expensive). For the general problem in AI agents, *see also* STUART RUSSELL & PETER NORVIG, *ARTIFICIAL INTELLIGENCE: A MODERN APPROACH* § 2.4.1 (4th ed. 2020) (demonstrating why the table-driven approach to agent design—specifying in advance the correct action for every possible percept sequence—is unworkable: even for chess, the lookup table would need more entries than the observable universe has atoms, and no designer would have time to construct it).

⁴²This is the operational consequence of the problem described *supra* note 41. *See* HART, *supra* note 41, at 77–79 (analyzing how parties fill contractual gaps through renegotiation and adaptation when complete contingent contracting is infeasible).

⁴³Convention-formation through mutual adaptation has been explored extensively in game theory. *See, e.g.*, THOMAS C. SCHELLING, *THE STRATEGY OF CONFLICT* 54–80 (1980) (discussing the role of focal points and tacit coordination in solving problems without direct communication). Contemporary studies extend this account: decentralized populations converge on shared conventions and display directional opinion dynamics driven by asymmetric persuasion rather than indiscriminate agreement. *See* Ashery et al., *supra* note 4, at 3; Erica Cau, Valentina Pansanella, Dino Pedreschi & Giulio Rossetti, *Selective Agreement, Not Sycophancy: Investigating Opinion Dynamics in LLM Interactions*, 14 EPJ DATA SCI. art. 59, at 18–19 (2025), <https://doi.org/10.1140/epjds/s13688-025-00579-1> (extending the account to explicit multi-agent debate and finding convergence via structured, asymmetric persuasion and measurable influence of logical fallacies).

⁴⁴*See* U.C.C. § 1-303(b)–(c) (AM. L. INST. & UNIF. L. COMM'N 2022) (defining “course of dealing” and “usage of trade”).

⁴⁵*See* U.C.C. § 1-303(d) (AM. L. INST. & UNIF. L. COMM'N 2022) (providing that a course of performance, course of dealing, or usage of trade “may give particular meaning to specific terms of the agreement, and may supplement or qualify the terms of the agreement”); *see also* Lisa Bernstein, *Merchant Law in a Merchant Court: Rethinking the Code's Search for Immanent Business Norms*, 144 U. PA. L. REV. 1765, 1789, 1799 (1996) (documenting that transactors leave aspects of their relationships to the extralegal realm where “the transaction costs of including a sufficiently well-specified written provision in their contract would exceed the benefits,” including an unwritten weighing practice that yields “significant transaction cost savings”). *But see id.* at 1768, 1802, 1805–06 (challenging the premise that courts should discover and apply immanent business norms, on the grounds that the relationship-preserving norms transactors follow in cooperation are not the end-game norms they would want a tribunal to apply, and that judicial resort to unwritten usage generates interpretive uncertainty rather than predictability). The doctrine proposed here meets that concern by conditioning recognition on consistency with express terms and confining it to interpretive, calibrated effect. *See infra* Part III.B–C; *see also infra* note 214 and accompanying text (engaging the critique directly).

decades, algorithmic practices can emerge, stabilize, and dissolve within hours, driven by the iterative learning of agents operating at machine speed.⁴⁶ This compressed timescale challenges legal frameworks built for human temporality, even as the practices themselves are born of the same functional necessity: coordination in the face of uncertainty.

B. A Taxonomy of Emergent Practices

The MAS literature studies practice emergence as a dynamic process of creation, diffusion, and stabilization, shaped by network structure, enforcement mechanisms, and the underlying coordination problem.⁴⁷ In contrast, the law must decide when an emergent practice should carry legal consequence—for those who transact through the agents or are affected by them, and for the principals who design, deploy, answer for, and ultimately benefit from them. That decision turns on notice, reliance, accountability, significance, and proof, bounded by contract and by public law; it is an inquiry the technical literature has no occasion to undertake.

We organize emergent practices along five *dimensions*—*Prevalence* (how widespread and long-lasting a practice is), *Function* (what problem it solves), *Legibility* (whether it is visible to others and can be shown from system records), *Genesis & Governance* (how it was formed and who shaped it), and *Mechanisms & Enforcement* (what keeps it in place). Together, these dimensions clarify *what* pattern is at issue, *how* it operates, and *whether* it warrants legal recognition; they provide the analytical foundation for the F-NORM test proposed in Part III.

A final characteristic cuts across these five dimensions: whether the coordination a practice

⁴⁶Cf. Ashery et al., *supra* note 4, at 3 (documenting decentralized convention formation arising through repeated interactions in LLM populations); *see supra* note 32 (millisecond-scale response budgets). At machine interaction rates, the round counts these studies report correspond to minutes or hours of wall-clock time.

⁴⁷For surveys of emergent practices in MAS, *see generally* Bastin Tony Roy Savarimuthu & Stephen Cranefield, *Norm Creation, Spreading and Emergence: A Survey of Simulation Models of Norms in Multi-Agent Systems*, 7 MULTIAGENT & GRID SYS. 21, 27–30 (2011) (proposing a five-phase norm life-cycle—creation, identification, spreading, enforcement, and emergence—and treating interaction topology as bearing on “all phases of norm development”); Christopher K. Frantz & Gabriella Pigozzi, *Modeling Norm Dynamics in Multi-Agent Systems*, 5 J. APPLIED LOGICS—IfCoLog J. LOGICS & THEIR APPL. 491, 522–29 (2018) (proposing a norm life-cycle in which emergence is “a result of iterative Transmission, Identification, Internalization and Enforcement processes”); Andreea Morris-Martin, Marina De Vos & Julian Padget, *Norm Emergence in Multiagent Systems: A Viewpoint Paper*, 33 AUTON. AGENTS & MULTI-AGENT Sys. 706, 716–17 (2019) (comparing three-, four-, and five-stage life-cycle models).

achieves is benign, or whether it produces unlawful or socially harmful outcomes. Emergent practices arise because coordination benefits the agents that achieve it. But such benefit does not make the resulting pattern fair to others, efficient overall, or lawful. Research demonstrates that coordinated equilibria can serve narrow interests or encode systematic harms.⁴⁸ Examples of such *harmful equilibria* include exclusionary routing protocols, collusive pricing rhythms, and discriminatory resource allocation patterns. Barring practices that violate or facilitate the violation of mandatory public law is the work of the Legality Screen, developed in Part III.B. We take the dimensions in turn, and summarize them in Table 1.

1. *Prevalence (Scope & Duration).*

Prevalence captures how far a practice spreads (*scope*) and how long it holds within the relevant interaction window (*duration*). Scope ranges from practices localized among a few agents to those adopted population-wide.⁴⁹ We identify three scales, adapting the literature’s local-versus-global framework for legal analysis:⁵⁰

- *Micro-Practices*: Highly localized practices governing interactions between a limited set of agents. An example is a queuing protocol between two agents contending for the same resource.
- *Meso-Practices*: Broader practices covering an entire digital marketplace or platform. An example is a shared bidding cadence in an advertising auction, adopted by many independent agents.

⁴⁸See Franks et al., *infra* note 65, at 350 (finding that “small proportions of agents in a population are able to significantly manipulate the conventions that are adopted by relatively large societies,” and that “malicious or faulty agents can disrupt convention emergence with ease”); Ashery et al., *supra* note 4, at 6–7 (finding that social coordination “can give rise to collective biases” not deducible from analyzing agents in isolation, and that “a minority of committed agents can impose their preferred convention on a majority settled on a different one”).

⁴⁹See Onkur Sen & Sandip Sen, *Effects of Social Network Topology and Options on Norm Emergence*, in COORDINATION, ORGS., INSTS. & NORMS IN AGENT SYS. V 211, 215–20 (Julian Padget et al. eds., 2010); Chao Yu, Minjie Zhang, Fenghui Ren & Xudong Luo, *Emergence of Social Norms through Collective Learning in Networked Agent Societies*, in PROC. 12TH INT’L CONF. ON AUTON. AGENTS & MULTIAGENT SYS. 475, 478–81 (2013).

⁵⁰See Morris-Martin et al., *supra* note 47, at 716.

- *Macro-Practices*: Widespread practices spanning an entire industry or ecosystem, often involving agents from many different platforms. This is the closest algorithmic analogue to traditional “usage of trade.”⁵¹

Duration tracks a practice’s lifecycle from emergence through stabilization to potential decay.⁵² Transient practices emerge to solve immediate coordination problems and then dissipate; persistent practices stabilize through repeated reinforcement and become enduring features of the operational environment.

2. *Function (Problem Solved).*

Function identifies which of three problems a practice solves—coordination, cooperation, or fairness:⁵³

- *Conventional practices* solve pure coordination problems—situations where agents need to align on one choice (e.g., which side of the corridor to use, what bidding cadence to adopt). These equilibria are self-enforcing: once established, coordination itself provides the reward.⁵⁴
- *Essential practices* solve cooperation problems where individual incentives favor defection but collective welfare requires cooperation (e.g., resource-sharing, contribution to common infrastructure). These typically depend on mechanisms that alter the incentive structure,

⁵¹Cf. U.C.C. § 1-303(c) (AM. L. INST. & UNIF. L. COMM’N 2022) (defining “usage of trade” as a practice with regularity of observance “in a place, vocation, or trade”). See *infra* Part II.A (analyzing how the U.C.C.’s place-, vocation-, and trade-based scope assumptions strain when applied to algorithmic emergent practices).

⁵²See Frantz & Pigozzi, *supra* note 47, at 518–25; cf. Savarimuthu & Cranefield, *supra* note 47, at 21, 27–30 (five-phase life-cycle model ending at emergence, without a decay stage).

⁵³See, e.g., Young, *supra* note 3, at 59–60 (analyzing coordination games); ROBERT AXELROD, THE EVOLUTION OF COOPERATION ch. 1 (rev. ed. 2006) (1984) (analyzing cooperation dilemmas). For MAS applications, see Yu et al., *supra* note 49, at 478, 480 (evaluating norm-convergence efficiency in coordination games as action sets vary); on the cooperation side, see Villatoro et al., *infra* note 55, at 415–17 (dynamic sanctioning sustaining compliance in cooperation dilemmas).

⁵⁴See Young, *supra* note 3, at 57 (defining conventions as customary, expected, and self-enforcing patterns to which everyone conforms because everyone else does); SCHELLING, *supra* note 43, at 54–62 (explaining coordination as mutually reinforcing without external enforcement).

such as reciprocity in repeated play, sanctions that penalize violations while signaling the practice they enforce, and metapractices that oblige agents to punish.⁵⁵

- *Prosocial practices* address fairness and welfare concerns that extend beyond simple payoff maximization, such as preventing starvation, reducing inequity, or protecting vulnerable agents.⁵⁶

The three categories differ in what they produce: conventional practices, alignment (throughput, reduced conflicts); essential practices, compliance and sanctioning records (adherence rates, penalty events); prosocial practices, distributional effects (variance in agent welfare or waiting times).⁵⁷

3. Legibility (*Observability & Detection*).

Legibility addresses whether a practice is encoded as a rule anywhere in the system and whether it can be proved from behavioral records. The literature distinguishes *explicit practices*, in which the governing rule is represented as such—whether in an agent’s internal state or in shared artifacts such as protocol flags or documented policies—from *implicit practices*, in which no represented rule exists and the regularity must be inferred from observed patterns of interaction.⁵⁸ Detecting them is the subject of a substantial body of work on *practice-identification algorithms*: computational

⁵⁵ See Daniel Villatoro, Giulia Andrighetto, Jordi Sabater-Mir & Rosaria Conte, *Dynamic Sanctioning for Robust and Cost-Efficient Norm Compliance*, in PROC. 22D INT’L JOINT CONF. ON ARTIF. INTELL. 414, 415–17 (2011); Samhar Mahmoud, Nathan Griffiths, Jeroen Keppens & Michael Luck, *Establishing Norms with Metanorms over Interaction Topologies*, 31 AUTON. AGENTS & MULTI-AGENT SYS. 1344, 1345 (2017). On the term, see Airiau et al., *supra* note 4, at 780 (distinguishing conventions, which solve coordination problems, from essential norms, where individual and collective interests conflict).

⁵⁶ See, e.g., Gary E. Bolton & Axel Ockenfels, *ERC: A Theory of Equity, Reciprocity, and Competition*, 90 AM. ECON. REV. 166, 171–72 (2000) (modeling inequity aversion); JOHN RAWLS, A THEORY OF JUSTICE §§ 13, 26, at 72, 132–33 (rev. ed. 1999) (1971) (stating the difference principle and the two principles’ relation to the maximin rule). For a MAS implementation of inequity aversion and Rawlsian fairness, see Mehdi Mashayekhi, Nirav Ajmeri, George F. List & Munindar P. Singh, *Prosocial Norm Emergence in Multi-agent Systems*, 17 ACM TRANS. AUTON. & ADAPT. SYS. art. 3, 3:1, 3:3–3:4 (Sept. 2022), <https://doi.org/10.1145/3540202> (implementing inequity aversion as a guilt disutility and adopting the Rawlsian maximin criterion as the fairness metric in a decentralized norm-emergence framework).

⁵⁷ For the corresponding metrics, see Yu et al., *supra* note 49, at 478–79 (convergence ratios); Villatoro et al., *supra* note 55, at 418–19 (cooperation, resilience, and enforcement-cost measures); Mashayekhi et al., *supra* note 56, at 3:15 (Gini-based inequity comparisons).

⁵⁸ Morris-Martin et al., *supra* note 47, at 712–13 (distinguishing explicitly represented norms from implicit ones, in which agents follow predetermined strategies without any express representation).

methods that reconstruct the regularity from interaction traces through association-rule mining or pattern analysis.⁵⁹

4. *Genesis & Governance (Source & Locus).*

Genesis and governance identify where a practice originates and who shapes it. The literature distinguishes *prescriptive* (top-down, authority-imposed) from *emergent* (bottom-up, peer-generated) practices, recognizing that many systems exhibit hybrid dynamics in which institutional rules channel or constrain otherwise decentralized emergence.⁶⁰ For legal analysis, we refine this along three loci of governance:

- *Bilateral practices* are shaped by the two parties themselves, arising as a course of performance within a single agreement or as a course of dealing across their prior transactions.⁶¹
- *Multilateral practices* are shaped by no single participant: they emerge from decentralized interactions among many peers (classic bottom-up emergence).
- *Platform-mediated practices* are shaped—though not always explicitly dictated—by an intermediary’s rules, incentive structures, or architectural constraints (e.g., rate limits, priority queues, default configurations).

The loci differ in whom they reach: bilateral practices trace to the parties themselves; multilateral practices may lack a single responsible actor; platform-mediated practices implicate the platform operator’s design choices.⁶² These loci are independent of scale: a platform-mediated

⁵⁹ See Tony Savarimuthu, Stephen Cranefield, Maryam A. Purvis & Martin K. Purvis, *Obligation Norm Identification in Agent Societies*, 13 J. ARTIF. SOC’YS & SOC. SIM. art. 3, ¶ 4.31 (Oct. 2010), <https://www.jasss.org/13/4/3.html> [<https://perma.cc/6VA7-AVBD>] [hereinafter Savarimuthu et al., *Obligation Norms*]; Bastin Tony Roy Savarimuthu, Stephen Cranefield, Maryam A. Purvis & Martin K. Purvis, *Identifying Prohibition Norms in Agent Societies*, 21 A.I. & L. 1, 19–22 (2013) [hereinafter Savarimuthu et al., *Prohibition Norms*]. See also *infra* Part I.C (cataloguing the record classes); *infra* Part I.D (detailing how each supports the F-NORM criteria).

⁶⁰ See Morris-Martin et al., *supra* note 47, at 712–13 (contrasting prescriptive and emergent approaches); Andrés García-Camino, Juan A. Rodríguez-Aguilar, Carles Sierra & Wamberto Vasconcelos, *Constraint Rule-Based Programming of Norms for Electronic Institutions*, 18 AUTON. AGENTS & MULTI-AGENT SYS. 186, 195, 199–202 (2009) (treating institutional rules as normative constraints).

⁶¹ Cf. U.C.C. § 1-303(a)–(b) (AM. L. INST. & UNIF. L. COMM’N 2022). See *infra* Part II.A (analyzing how the U.C.C.’s practice hierarchy applies to algorithmic emergent practices).

⁶² See *infra* Part III.B (mapping this dimension to the Attribution factor).

practice may be micro-, meso-, or macro-scale, and a macro-scale practice may arise with no platform involvement at all.

5. *Mechanisms & Enforcement (Structure & Incentives).*

Mechanisms and enforcement identify what keeps a practice in place: the network *structure* through which it spreads and the *incentives* that sustain it. The literature treats both as first-class determinants of practice stability, not peripheral factors.

Network structure—the topology of connections among agents—shapes how far and how fast a practice propagates. Dense or small-world topologies accelerate convergence; sparse or fragmented networks trap practices in local clusters; scale-free networks let hub agents block a practice from establishing itself altogether. Metrics such as clustering coefficient, degree distribution, and path length therefore bear directly on whether a practice remains micro-scale or reaches meso- or macro-scale.⁶³

Enforcement mechanisms stabilize practices by aligning individual incentives with collective expectations. Punishment deters by imposing costs on violators; sanctioning pairs that cost with a signal identifying the practice being defended, and so sustains cooperation more effectively and at lower cost. Metapractices extend enforcement one level up—obliging agents to punish those who fail to punish—making compliance individually rational even in cooperation dilemmas.⁶⁴

Enforcement is not the only way a practice can be shaped. Research on *influencer agents* demonstrates that small groups or central actors can deliberately steer emergence toward specific equilibria—including, when the steering minority is adversarial, toward low-quality conventions. That capacity yields coordination benefits, but its exercise can also ripen into the exclusionary or collusive practices this Article takes up in Part IV.⁶⁵

⁶³See Yu et al., *supra* note 49, at 476–77, 480 (path-length, degree, and clustering effects on convergence); Mahmoud et al., *supra* note 55, at 1344 (norm establishment achievable in lattices and small worlds but not scale-free networks, “due to the problematic effects of hubs”); Morris-Martin et al., *supra* note 47, at 716 (community network structure yielding local rather than global emergence).

⁶⁴The MAS literature calls these obligations *metanorms*. See Villatoro et al., *supra* note 55, at 414–15, 418–19; Mahmoud et al., *supra* note 55, at 1348–49.

⁶⁵Henry Franks, Nathan Griffiths & Arshad Jhumka, *Manipulating Convention Emergence Using Influencer Agents*, 26

Table 1: Analytical Dimensions Adapted from the MAS Literature

Dimension	Core Questions	Examples	Sources
Prevalence (Scope & Duration)	How widely does the practice spread? How long does it persist within the relevant interaction window?	Local cluster conventions vs. population-wide adoption; transient coordination spikes vs. persistent routines.	1, 2, 3, 6, 13
Function (Problem Solved)	What problem does the practice solve: coordination, cooperation, or fairness?	Driving-side conventions (coordination); resource contribution with sanctions (cooperation); anti-starvation protocols (fairness).	3, 4, 5, 12, 14
Legibility (Observability & Detection)	Is the practice explicitly represented or implicit? Can it be detected from behavioral traces?	Explicitly represented rules vs. implicit converged policies; practice identification from interaction logs.	6, 7, 8
Genesis & Governance (Source & Locus)	Is the origin top-down or bottom-up? Is the locus bilateral, multilateral, or platform-mediated?	Authority-prescribed rules vs. peer-generated conventions; platform rate limits shaping behavior.	6, 9
Mechanisms & Enforcement (Structure & Incentives)	What topology enables propagation? What mechanisms sustain compliance?	Dense networks accelerating convergence; sanctions and metapractices enforcing cooperation; influencer agents steering emergence.	2, 3, 5, 6, 10, 11

Sources: 1. Savarimuthu & Cranefield, *supra* note 47. 2. Sen & Sen, *supra* note 49. 3. Yu et al., *supra* note 49. 4. Young, *supra* note 3. 5. Villatoro et al., *supra* note 55. 6. Morris-Martin et al., *supra* note 47. 7. Savarimuthu et al., *Obligation Norms*, *supra* note 59. 8. Savarimuthu et al., *Prohibition Norms*, *supra* note 59. 9. García-Camino et al., *supra* note 60. 10. Mahmoud et al., *supra* note 55. 11. Franks et al., *supra* note 65. 12. Mashayekhi et al., *supra* note 56. 13. Frantz & Pigozzi, *supra* note 47. 14. AXELROD, *supra* note 53.

C. The Evidentiary Substrate

Courts often evaluate machine-generated evidence by looking not at the machine itself but at its auditable *outputs and processes*.⁶⁶ A similar approach can be developed for AI agents. By *capturing* the right records, *authenticating* and *preserving* them, *analyzing* them for regularity and effect, and *reproducing* core findings for a neutral to verify, we can characterize an emergent practice—its existence, scope, operation, and impact—and apply the relevant doctrinal tests.

1. *Capture.*

While an agent’s *internal logic* may be opaque, its *external actions* are typically recorded in meticulous detail—and because the interactions are natively digital, the record arrives as timestamped, machine-readable traces of a kind rarely available for human conduct. Capture is therefore largely a matter of collecting what the system already writes down in the ordinary course of its operation. Proof runs on that record, defusing the “black-box” problem without ever opening the box.⁶⁷

2. *Authenticate and Preserve.*

Two routes to authenticating machine-generated records are canonical. First, FED. R. EVID. 901(b)(9) admits evidence of outputs from a described process or system upon a showing that it “produces an accurate result.” Second, FED. R. EVID. 902(13)–(14) allow for the *self-authentication* of electronic records via a qualified person’s certification, reducing the need for live custodians.⁶⁸

⁶⁶See, e.g., FED. R. EVID. 901(b)(9) (authenticating the result of a process or system by evidence describing it and showing that it produces an accurate result); FED. R. EVID. 902(13)–(14) (self-authentication of electronic records); FED. R. EVID. 803(6) (business records). Habit evidence offers a further route where a routine is attributable to the deploying organization. Cf. FED. R. EVID. 406 (routine practice of an organization); see *infra* note 106 and accompanying text.

⁶⁷See, e.g., FRANK PASQUALE, *THE BLACK BOX SOCIETY: THE SECRET ALGORITHMS THAT CONTROL MONEY AND INFORMATION* 3, 8 (2015) (defining a “black box” as “a system whose workings are mysterious; we can observe its inputs and outputs, but we cannot tell how one becomes the other,” and observing that “authority is increasingly expressed algorithmically,” such that “[t]he values and prerogatives that the encoded rules enact are hidden within black boxes”).

⁶⁸See FED. R. EVID. 902(13)–(14). The business-records exception, FED. R. EVID. 803(6), also commonly fits routinely

Digital-forensics standards govern the chain of custody,⁶⁹ and parties can go further by sealing log bundles with cryptographic timestamps, recording digests in append-only transparency logs, and attesting to the integrity of the environment in which the agent ran.⁷⁰ Once a duty to preserve attaches, the loss of ephemeral traces risks sanctions under FED. R. CIV. P. 37(e), and the ordinary proportionality and form-of-production rules govern what must be produced and how.⁷¹

3. *Analyze.*

To support recognition, the analysis must (i) *identify* a stable behavioral regularity within the defined interaction window (Regularity) and (ii) *show its effect* (Materiality) with court-familiar causal tools. Practice-identification methods from the MAS literature surface the regularity itself, as do process- and sequence-mining techniques.⁷² Event-study and interrupted-time-series designs then quantify the outcome shifts when a practice emerges or is disrupted.⁷³ Summaries and charts

kept logs and telemetry. Courts have long applied the authentication and business-records rules to machine-generated records, and the 2017 addition of Rules 902(13)–(14) supplied a certification route. *See, e.g.,* Lorraine v. Markel Am. Ins. Co., 241 F.R.D. 534, 538, 542–52, 570–74 (D. Md. 2007) (providing a comprehensive roadmap for the admissibility of electronically stored information); *United States v. Lizarraga-Tirado*, 789 F.3d 1107, 1109–10 (9th Cir. 2015) (holding that a Google Earth satellite image bearing an automatically generated GPS tack was not hearsay because the assertion was made by a machine rather than a person, and explaining that machine-reliability concerns “are addressed by the rules of authentication,” including FED. R. EVID. 901(b)(9)).

⁶⁹*See, e.g.,* INT’L ORG. FOR STANDARDIZATION & INT’L ELECTROTECHNICAL COMM’N, ISO/IEC 27037:2012, INFORMATION TECHNOLOGY — SECURITY TECHNIQUES — GUIDELINES FOR IDENTIFICATION, COLLECTION, ACQUISITION AND PRESERVATION OF DIGITAL EVIDENCE (2012).

⁷⁰*See* INTERNET ENG’G TASK FORCE (IETF), RFC 3161, INTERNET X.509 PUBLIC KEY INFRASTRUCTURE TIME-STAMP PROTOCOL (TSP) (2001); IETF, RFC 9162, CERTIFICATE TRANSPARENCY VERSION 2.0 (2021); IETF, RFC 9334, REMOTE ATTESTATION PROCEDURES (RATS) ARCHITECTURE (2023).

⁷¹*See* FED. R. CIV. P. 37(e), 26(b)(1), 34(b)(1)(C), 34(b)(2)(E). The 2015 amendment to Rule 37(e) now supplies the governing framework for the loss of electronically stored information. Curative measures require a finding of prejudice; the severe measures—adverse-inference instructions, presumptions, or dismissal—require a finding of intent to deprive another party of the information’s use. FED. R. CIV. P. 37(e)(2). For the foundational duty to preserve electronic data, *see* *Zubulake v. UBS Warburg LLC*, 220 F.R.D. 212, 216–18 (S.D.N.Y. 2003) [hereinafter *Zubulake IV*]; on counsel’s duty to implement and monitor a litigation hold, *see* *Zubulake v. UBS Warburg LLC*, 229 F.R.D. 422, 431–34 (S.D.N.Y. 2004) [hereinafter *Zubulake V*]. *See also* THE SEDONA CONFERENCE, *The Sedona Principles, Third Edition: Best Practices, Recommendations & Principles for Addressing Electronic Document Production*, 19 SEDONA CONF. J. 1, 169, 174 (2018) (stating that production should be made “in the form or forms in which it is ordinarily maintained or that is reasonably usable given the nature of the electronically stored information and the proportional needs of the case,” and cautioning that a “reasonably usable” form “need not necessarily be its native format”).

⁷²*See* Savarimuthu et al., *Obligation Norms*, *supra* note 59, ¶ 4.31; Savarimuthu et al., *Prohibition Norms*, *supra* note 59, at 19–22; *see generally* WIL M.P. VAN DER AALST, *PROCESS MINING: DATA SCIENCE IN ACTION* (2d ed. 2016).

⁷³*See* A. Craig MacKinlay, *Event Studies in Economics and Finance*, 35 J. ECON. LITERATURE 13, 14–15 (1997) (setting

that condense voluminous electronically stored information (ESI) are admissible under FED. R. EVID. 1006,⁷⁴ and expert testimony on these analyses is screened under FED. R. EVID. 702, *Daubert*, and *Kumho Tire*.⁷⁵

4. *Reproduce.*

Verifiability means a neutral expert can reconstruct the routine—by deterministic (seeded) replay in a sandbox against pinned versions and configurations, and by working from the audit documentation that AI-governance frameworks already recommend.⁷⁶ Where needed, courts can use FED. R. CIV. P. 26(c) protective orders to permit targeted disclosures of proprietary data without loss of confidentiality.⁷⁷ These materials give a court the basis for deciding whether a claimed practice is independently reproducible and whether the expert analysis of it is reliable.⁷⁸

Three classes of data form the evidentiary substrate. *Primary data* are the high-frequency interaction traces—timestamped transaction logs, API calls, scheduler and queue events, message

out the standard event-study design: defining the event and the event window, estimating normal performance over a separate pre-event estimation window, and measuring the event’s impact as the abnormal outcome—the actual value over the event window minus the modeled normal value).

⁷⁴See FED. R. EVID. 1006 (as amended Dec. 1, 2024) (permitting summaries of voluminous admissible records “whether or not they have been introduced into evidence,” and requiring that the underlying originals or duplicates be made available to other parties); *cf.* FED. R. EVID. 107 (governing illustrative aids); *see also* United States v. Bray, 139 F.3d 1104, 1109–10 (6th Cir. 1998) (requiring that the underlying records be voluminous, independently admissible, and made available to the opposing party).

⁷⁵See FED. R. EVID. 702 (as amended Dec. 1, 2023); *Daubert v. Merrell Dow Pharm., Inc.*, 509 U.S. 579, 592–95 (1993); *Kumho Tire Co. v. Carmichael*, 526 U.S. 137, 147–49 (1999).

⁷⁶See NAT’L INST. OF STANDARDS & TECH., ARTIFICIAL INTELLIGENCE RISK MANAGEMENT FRAMEWORK (AI RMF 1.0) 15–16 (NIST AI 100-1, 2023), <https://doi.org/10.6028/NIST.AI.100-1> [<https://perma.cc/QX98-LABQ>] (grounding trustworthy AI in accountability and transparency); NAT’L INST. OF STANDARDS & TECH., NIST AI RMF PLAYBOOK, <https://airc.nist.gov/airmf-resources/playbook/> [<https://perma.cc/XX2K-FC68>] (last visited Aug. 2, 2026) (recommending transparency tools such as model cards and data statements).

⁷⁷See FED. R. CIV. P. 26(c)(1)(G) (authorizing protective orders requiring that a trade secret or other confidential research, development, or commercial information not be revealed or be revealed only in a specified way).

⁷⁸Proposed Federal Rule of Evidence 707 would subject machine-generated evidence offered without a sponsoring expert to Rule 702’s reliability standards, so that—whichever lane the proof travels—admissibility would turn on the reproducibility described here. *See* COMM. ON RULES OF PRAC. & PROC., JUD. CONF. OF THE U.S., PRELIMINARY DRAFT OF PROPOSED AMENDMENTS TO THE FEDERAL RULES OF APPELLATE, BANKRUPTCY, CIVIL, AND CRIMINAL PROCEDURE, AND THE FEDERAL RULES OF EVIDENCE 109–11 (Aug. 2025), https://www.uscourts.gov/sites/default/files/document/preliminary-draft-of-proposed-amendments-to-federal-rules_august2025.pdf [<https://perma.cc/5JT2-8E6Y>]. The Advisory Committee has since retained the proposal for further study. *See* REP. OF THE ADVISORY COMM. ON EVID. RULES 1 (May 17, 2026), https://www.uscourts.gov/sites/default/files/document/advisory_committee_on_evidence_rules_may_2026.pdf [<https://perma.cc/T3TW-YAWW>].

traces, and telemetry—that show what the agents did.⁷⁹ *Secondary data* provide the context needed to interpret those traces and attribute conduct: version-control histories, deployment manifests, signed configuration bundles, feature-flag and policy files, Software Bills of Materials (SBOMs), and reward or objective descriptors. *Tertiary data*—public exchange feeds, broadcast telemetry, or platform dashboards—supply independent cross-checks and, being public, evidence of what counterparties could see. Where a practice emerges from learned parameters alone—leaving no configuration diff, feature flag, or commit to discover—the secondary record still captures the legally salient choice: the decision to deploy, and to keep deploying, the system that learned it.

D. Characterizing Emergent Practices

The three classes of data characterize emergent practices along the five dimensions developed in Part I.B, and connect each dimension to the criteria of the F-NORM test developed in Part III.B:⁸⁰

- *Prevalence (Scope & Duration)*. Scope and duration are established from the *primary data*. Timestamped logs can demonstrate a practice’s stability and persistence within an interaction window (*duration*). Logs across a population of agents, and the topology of their interactions, potentially cross-referenced with public feeds and other *tertiary data*, can establish how widely the practice was adopted (*scope*).⁸¹ This dimension informs Regularity and Foreseeability: a practice that persists within an interaction window is a regularity rather than an episode, and one adopted population-wide is a practice counterparties had reason to anticipate.
- *Function (Problem Solved)*. A practice’s function is inferred from its effects in the *primary data*: event studies can show a reduction in conflicts (coordination), logs can reveal sanctioning

⁷⁹Standardized observability formats and event-log schemas make these data available at fine granularity. See, e.g., OPENTELEMETRY, <https://opentelemetry.io> [<https://perma.cc/GY9E-Z6CD>] (last visited Apr. 22, 2026) (a vendor-neutral, open-source observability framework). On log management, see NAT’L INST. OF STANDARDS & TECH., SPECIAL PUB. 800-92, GUIDE TO COMPUTER SECURITY LOG MANAGEMENT (2006), <https://doi.org/10.6028/NIST.SP.800-92> [<https://perma.cc/C3ZP-2LLG>].

⁸⁰The assignments are typical rather than exclusive.

⁸¹On the authentication and log-management anchors for these records, see *supra* notes 66, 68, and 79. On network topology as a determinant of scope, see Sen & Sen, *supra* note 49, at 217–20.

behavior (cooperation), and distributional metrics can show that no agent is perpetually denied a shared resource (fairness). A reward or objective descriptor in a configuration file—*secondary data*—can corroborate the inference by specifying what the agent was optimized to achieve. This dimension informs Materiality and the Legality Screen: the problem a practice solves is what makes it matter to the interaction, and evidence of what it accomplishes is also evidence of whether the law permits it.

- *Legibility (Observability & Detection)*. An *implicit* practice can be recovered from the *primary data* by statistical analysis; an *explicit* practice can be read from *secondary data* such as policy files or API documentation; and whether counterparties could observe either can be shown from *tertiary data* such as public dashboards.⁸² This dimension informs Foreseeability and Verifiability: what a counterparty could see is what it can fairly be charged with anticipating, and what a third party can reproduce is what a court can independently check.
- *Genesis & Governance (Source & Locus)*. A practice's origin and locus are traced through the *secondary data*: version-control histories and deployment manifests can pinpoint the software update that introduced a behavior—and with it the party whose choice put that behavior in the system—while configuration files can reveal whether a platform shaped the practice through its default settings or architectural constraints (*platform-mediated*).⁸³ This dimension informs Attribution and—where a platform's defaults are themselves fixed by its terms of service—the Consistency gate, which bars recognition of a practice that contradicts that same instrument.
- *Mechanisms & Enforcement (Structure & Incentives)*. The mechanisms that sustain a practice are read from the *primary data*, directly and structurally: logs may show an agent deprioritized after departing from the practice (direct evidence of enforcement), and the

⁸²Where verification will be contested, the record itself can be hardened: cryptographically sealed log bundles carry their own integrity proofs, *see supra* note 70, and replay packs pinning the software versions, seeds, and configurations permit independent re-execution subject to expert screening, *see supra* notes 76 and 75.

⁸³Software bills of materials and feature-flag records serve the same tracing function; on the handling standards for such artifacts, *see supra* note 69.

interaction topology reconstructed from those traces can show whether adherence depends on a few central hubs or is diffused across the population. This dimension informs Regularity and the Legality Screen: enforcement is what distinguishes a stable practice from a run of coincidences, and the same evidence can flag exclusionary or steering dynamics relevant to the Screen.

Having established what emergent practices are, how they form, and how they can be characterized from the evidentiary substrate, this Article now turns in Part II to the existing legal landscape, asking whether the doctrines courts already apply can accommodate these practices and, if not, where they fall short.

II. THE DOCTRINAL SCAFFOLDING: EXISTING LAW AND ITS LIMITS

Though novel in its application to AI agents, the doctrine of functional norms is assembled from the scaffolding by which courts have long given informal practices legal effect: the private law of contract and tort, the evidence rules that prove practice, statutory frameworks for electronic transactions, agency principles for attributing automated conduct, and the mandatory backstops of public law. From each, this Article draws what it needs: from contract, terms of recognition, a hierarchy of sources, and the commercial significance that makes a practice matter; from tort, evidentiary weight; from the evidence rules, a proof path; from electronic transactions law, validity for automated acts; from agency, routes of attribution; and from public law, boundaries. This Part takes each body of law in turn, asking what it lends and where it strains.

A. Contract's Recognition of Practice

Commercial law treats parties' actual practices as often the most reliable guide to their understanding of their agreement.⁸⁴ The Uniform Commercial Code (U.C.C.) codifies this principle by

⁸⁴The Code's practice provisions rest on the premise that "[e]very fact-pattern of common life, so far as the legal order can take it in, carries within itself its appropriate, natural rules, its right law." KARL N. LLEWELLYN, *THE COMMON LAW TRADITION: DECIDING APPEALS* 122 (1960) (quoting Levin Goldschmidt).

according legal weight to informal practices. The express terms of an agreement and any such practices are construed whenever reasonable as consistent with one another; when reconciliation is unreasonable, the Code ranks the sources—express terms prevail, followed by “course of performance” (conduct within a specific contract), “course of dealing” (conduct across prior contracts), and finally “usage of trade” (a practice observed with such regularity in a place, vocation, or trade as to justify an expectation that it will be observed).⁸⁵

The Code’s parol-evidence framework admits evidence of these practices, which courts may use to interpret, supplement, and in limited cases qualify written terms.⁸⁶ Beyond interpretation, a sustained course of performance is also relevant to show waiver or modification: a “no-oral-modification” clause forecloses the modification route, but conduct that is clear and relied upon can still operate as a waiver.⁸⁷

Two background doctrines supply a fairness backstop. First, every contract governed by the Code carries a non-waivable duty of good faith in performance and enforcement.⁸⁸ That duty polices opportunism: an emergent practice cannot be invoked as cover for unfair dealing. Second, *unconscionability* permits courts to refuse enforcement of a contract or clause that is oppressively

⁸⁵ See U.C.C. § 1-303(e) (AM. L. INST. & UNIF. L. COMM’N 2022); *id.* § 1-303(c) (defining “usage of trade”). The RESTATEMENT (SECOND) OF CONTRACTS adopts a similar framework, adding that a “usage of trade” may include “a system of rules regularly observed even though particular rules are changed from time to time.” See RESTATEMENT (SECOND) OF CONTRACTS § 222(1) (AM. L. INST. 1981); *id.* § 223. See also *id.* § 203(b) (stating the parallel ranking).

⁸⁶ See U.C.C. § 2-202 (AM. L. INST. & UNIF. L. COMM’N 2022) (permitting explanation or supplementation by course of performance, course of dealing, or usage of trade); *id.* § 2-202 cmt. 2 (explaining that, “[u]nless carefully negated,” prior dealings and usages “have become an element of the meaning of the words used”). See also *id.* § 1-303(d) (providing that a course of performance, course of dealing, or usage of trade “may supplement or qualify the terms of the agreement”).

⁸⁷ See U.C.C. § 1-303(f) (AM. L. INST. & UNIF. L. COMM’N 2022) (a course of performance “is relevant to show a waiver or modification of any term inconsistent with the course of performance”); *id.* § 2-209(2), (4)–(5); *Wis. Knife Works v. Nat’l Metal Crafters*, 781 F.2d 1280, 1286–87 (7th Cir. 1986) (Posner, J.) (reversing and remanding for a new trial on waiver where the jury had not been instructed that an attempted oral modification barred by a no-oral-modification clause “can operate as a waiver” under § 2-209(4) only if the other party relied on it).

⁸⁸ See U.C.C. § 1-304 (AM. L. INST. & UNIF. L. COMM’N 2022); *id.* §§ 1-201(b)(20), 1-302(b) (defining “good faith” as “honesty in fact and the observance of reasonable commercial standards of fair dealing,” and barring disclaimer of the obligation while permitting the parties to set not-manifestly-unreasonable standards for measuring its performance); RESTATEMENT (SECOND) OF CONTRACTS § 205 (AM. L. INST. 1981). The Code extends these policies to usage itself: its good-faith and unconscionability provisions “apply to implicit clauses that rest on usage of trade,” and the courts’ policing of usage continues. U.C.C. § 1-303 cmt. 5 (AM. L. INST. & UNIF. L. COMM’N 2022); see *infra* Part III.B (developing the Legality Screen, which performs this policing function for algorithmic practices).

one-sided—a salient safeguard in digital markets built on standard-form adhesion contracts.⁸⁹

Courts have used these practice provisions to look beyond the four corners of a document to the commercial context. In *Nanakuli Paving & Rock Co. v. Shell Oil Co.*, for example, the Ninth Circuit vacated a judgment notwithstanding the verdict and reinstated an award resting on a deeply embedded trade usage of price protection. The court held that the jury could reasonably construe that usage as consistent with the express price term of a written asphalt supply contract.⁹⁰ Its emphasis fell on significance as much as consistency: price protection was a “realistic necessity” of the local market, “vital” to the buyer’s ability to win large government contracts—the concern the Materiality factor carries forward.⁹¹

Moreover, these principles are not confined to the sale of goods. The reach of the U.C.C.’s Article 2 is growing: long limited to transactions in “goods,” it now extends, under the 2022 amendments, to the sale-of-goods aspects of “hybrid transactions” that mix goods with the services, data, and software licenses typical of algorithmic commerce.⁹² And where a transaction falls outside Article 2 altogether, courts may still look to U.C.C. principles by analogy; the common law supplies parallel rules.⁹³

Yet for all this reach, three points of friction limit these doctrines’ application to AI agents.

The first friction is the problem of *transience*. The obstacle is not that the Code requires age—it does not. Official Comment 4 to § 1-303 abandons “[t]he ancient English tests for ‘custom,’” disclaims any requirement that a usage be “ancient or immemorial” or “universal,” and confirms

⁸⁹ See U.C.C. § 2-302 (AM. L. INST. & UNIF. L. COMM’N 2022); see also RESTATEMENT (SECOND) OF CONTRACTS § 208 (AM. L. INST. 1981) (stating the parallel rule for contracts generally).

⁹⁰ See 664 F.2d 772, 777, 780, 790–96 (9th Cir. 1981); see also *Columbia Nitrogen Corp. v. Royster Co.*, 451 F.2d 3, 8–10 (4th Cir. 1971) (holding that evidence of usage of trade and course of dealing was admissible to explain a contract’s price and quantity terms, without any precondition that the writing be found ambiguous).

⁹¹ See *Nanakuli*, 664 F.2d at 780; see *infra* Part III.B.

⁹² See U.C.C. § 2-102(1)–(2) (AM. L. INST. & UNIF. L. COMM’N 2022) (applying Article 2 to transactions in goods and, in a “hybrid transaction,” to its sale-of-goods aspects—or, where those aspects predominate, to the transaction as a whole). Before the 2022 amendments, the section reached “transactions in goods” alone. See U.C.C. § 2-102 (AM. L. INST. & UNIF. L. COMM’N 2018).

⁹³ See RESTATEMENT (SECOND) OF CONTRACTS §§ 219–223 (AM. L. INST. 1981) (governing how usage of trade and course of dealing give meaning to, supplement, or qualify an agreement); *id.* § 202(4) (any course of performance “accepted or acquiesced in without objection is given great weight in the interpretation of the agreement”).

that “full recognition is thus available for new usages.”⁹⁴ The obstacle is calibration. “Regularity of observance” is calibrated to human transactional rhythms, in which a pattern accretes across dealings separated by days or months. A practice that emerges, stabilizes, and dissolves within a single interaction window in a high-speed environment forces a question that the Code’s text never answers: what quantum of repetition, over what interval, counts as regular at machine speed.

Nor can course of performance absorb the problem. That doctrine reaches repeated conduct between the same parties within a particular transaction, while emergent practices arise across many agents and counterparties at once. Duration is not idle in the law of usage. It is the channel through which participants come to know a practice and the period over which a record of it accumulates. A usage observed for years can be presumed known in the trade and proved from ample evidence, but a practice that held for only two minutes leaves open the questions—who can fairly be charged with knowledge of it, and what record survives to establish it.

Second is the problem of *opacity*. The Code’s practice doctrines run on human awareness: a usage of trade binds parties who “are or should be aware” of it, and its proponent must give the other party notice “sufficient to prevent unfair surprise.”⁹⁵ Both rules strain when the “parties” are algorithms. The pattern itself is legible in the logs, but only to a forensic audit, and what the audit yields is conduct without rationale: the record shows what the agents did, not why they did it. A counterparty can therefore see the regularity and still be unable to state the rule it instantiates—and a rule that cannot be stated cannot be anticipated, which is what notice is meant to secure. Transience compounds the problem: the two questions it leaves open—knowledge and proof—both run through legibility. A usage that holds for years is legible in two ways—the data accumulate, and the humans in the trade who live with the practice observe it, discuss it, and come to anticipate it. A practice that forms and dissolves within a single interaction window is legible

⁹⁴ See U.C.C. § 1-303(c) (AM. L. INST. & UNIF. L. COMM’N 2022) (defining “usage of trade” by “such regularity of observance . . . as to justify an expectation that it will be observed”); *id.* § 1-303 cmt. 4; *see also* *Nanakuli*, 664 F.2d at 792 (quoting the comment’s abandonment of “[t]he ancient English tests for ‘custom’ ”); *supra* note 25.

⁹⁵ See U.C.C. § 1-303(d) (AM. L. INST. & UNIF. L. COMM’N 2022); *id.* § 1-303(g).

in neither: no community absorbs it, and there is little time for anyone to analyze it, foresee it, or object to it.⁹⁶

The third friction is *nonhuman origin*. The Code's locus categories are capacious: a usage may be "general to trade" or "particular to a special branch of trade," and *Nanakuli* read the statute's disjunctive "or" to bind parties to a usage of the place where they did business even though it was not the usage of their own vocation or trade.⁹⁷ Every category the Code offers is a form of human association: where people do business (a place), what calling they practice (a vocation), what market they serve (a trade). They answer two questions a court cannot avoid: which community is the relevant one, and who counts as a member. The affiliations they name are few, stable, and knowable—a merchant knows its own, and a court can find them. But an emergent practice can arise wholly among the agents using a particular software library or transacting on a single platform—a community bounded by protocol adoption and API access rather than by geography, calling, or market. Protocol membership has none of those properties: an agent may enter or leave at will, and a single firm's agents may inhabit many overlapping communities at once. A platform can be called a "place," but a venue is not an affiliation. Neither question finds an answer.

From contract, then, we borrow the Code's core principles and their limit: regularity of observance, recalibrated to the interaction window; awareness and notice, adapted into constructive, technical form as the Foreseeability factor; the commercial significance of a practice, carried forward as the Materiality factor; and the primacy of express terms, preserved in the Consistency gate.⁹⁸

⁹⁶Nor is the difficulty peculiar to the Code. International sales law runs on the same presupposition: the CISG binds parties to usages they "knew or ought to have known" and that are "widely known to, and regularly observed by, parties to contracts of the type involved"—knowledge and observation that presuppose humanly legible practices. See United Nations Convention on Contracts for the International Sale of Goods art. 9(2), Apr. 11, 1980, S. Treaty Doc. No. 98-9 (1983), 1489 U.N.T.S. 3. Even the Convention's most permissive channel—article 9(1), which binds parties to "practices which they have established between themselves"—presupposes parties who know their own practices, an assumption emergent machine behavior defeats.

⁹⁷See U.C.C. § 1-303(c) (AM. L. INST. & UNIF. L. COMM'N 2022) (defining "usage of trade" by reference to "a place, vocation, or trade"); *id.* § 1-303 cmt. 6; 664 F.2d at 791.

⁹⁸See *infra* Part III.B (developing these adaptations); see also *supra* note 87 (the waiver framework and its reliance condition).

B. Tort's Flexible Standard of Care

Tort law provides a complementary framework through its flexible treatment of custom: proof of a prevailing custom routinely informs what a reasonable person would do under the circumstances.⁹⁹ This framework, however, strains at the same joints as the Code's. The Restatement locates custom in "the custom of the community, or of others in like circumstances." At its strongest, a custom "establishes the standard by which those engaging in the activity assume they are bound."¹⁰⁰ That language rests on a human-association premise: for a practice that exists only among agents running one software library or trading on one platform, the relevant community and activity have no settled referent. And under New York's classic formulation, custom binds where it is "fairly well defined and in the same calling or business" so that the actor "may be charged with knowledge of it or negligent ignorance."¹⁰¹ That language rests on a human-awareness premise: knowledge and negligent ignorance are predicates that the case law developed for human actors, and a short-lived practice legible only in logs gives even the humans behind the agents little opportunity to know or culpably ignore it. These two premises track the Code's frictions: the association premise is the community problem of *nonhuman origin*, and the awareness premise is where *transience* and *opacity* converge. Custom's binding force runs through both.

The strain goes to identification, not weight: the rules that fix custom's weight ask nothing about who belongs or who knew. The treatment is evidentiary at its core. The foundational principle, articulated by Judge Learned Hand in *The T.J. Hooper*, is that custom is never a complete defense; an entire calling may be negligent.¹⁰² Custom cannot excuse the violation of a statute, ordinance, or administrative regulation. Where such an enactment prescribes a specific standard

⁹⁹ See RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 13 & cmts. a–b (AM. L. INST. 2010) (explaining that compliance with custom is evidence of nonnegligence but not conclusive).

¹⁰⁰ *Id.* § 13(a)–(b).

¹⁰¹ *Trimarco v. Klein*, 436 N.E.2d 502, 505–06 (N.Y. 1982).

¹⁰² See 60 F.2d 737, 740 (2d Cir. 1932) (declaring, in a case in which the court found "no custom at all" on the point, that "[i]ndeed in most cases reasonable prudence is in fact common prudence; but strictly it is never its measure; a whole calling may have unduly lagged in the adoption of new and available devices"); see also *Trimarco*, 436 N.E.2d at 505–06 (reaffirming that custom is relevant but not dispositive).

of conduct, an unexcused violation constitutes negligence per se, no matter how widespread the contrary practice.¹⁰³ That rule establishes a hard floor for the standard of care.¹⁰⁴ The floor, however, is not a ceiling: compliance with the enactment is evidence of due care but does not preclude a finding of negligence—above the floor, practice retains its evidentiary role.¹⁰⁵

Beyond the standard of care, the Federal Rules of Evidence supply an evidentiary hook. Rule 406 admits evidence of “an organization’s routine practice” to prove that on a particular occasion the organization acted in accordance with it. The rule’s categories—a person’s habit, an organization’s routine practice—do not name the machine; but a deployed system’s regularities, routed through the agency principles of Part II.D, are the routine practice of its deploying organization. When a firm’s agents almost always meet a specific stimulus with a specific response—initiating a jittered backoff protocol (pausing for a randomized interval before retrying), for example—that regularity is exactly the conduct the rule’s drafters described: a “regular practice of meeting a particular kind of situation with a specific type of conduct,” so ingrained it “may become semi-automatic.”¹⁰⁶ Neither of custom’s strained premises is implicated: the rule asks nothing about community membership and presumes no shared knowledge—it attaches to a single organization’s own regularity, the very thing interaction logs establish. What Rule 406 proves, though, is conformity—that the organization followed its routine on the occasion in question—not the existence or terms of a practice shared across parties; establishing that is the work of the evidentiary substrate detailed in Part I.C and the recognition test of Part III.

¹⁰³ See *Martin v. Herzog*, 228 N.Y. 164, 168, 126 N.E. 814 (1920) (Cardozo, J.) (“We think the unexcused omission of the statutory signals is more than some evidence of negligence. It is negligence in itself.”); RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 14 & cmt. a (AM. L. INST. 2010) (applying the rule to administrative regulations and local ordinances as well as statutes). See also *id.* § 15 cmt. a (“[N]or is it an excuse if there is a custom to depart from the statutory requirement.”).

¹⁰⁴ See *infra* Part III.B (developing the Legality Screen, which extends this principle by barring recognition of practices that violate—or facilitate the violation of—mandatory public law).

¹⁰⁵ See RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 16(a) (AM. L. INST. 2010) (“An actor’s compliance with a pertinent statute, while evidence of nonnegligence, does not preclude a finding that the actor is negligent under § 3 for failing to adopt precautions in addition to those mandated by the statute.”).

¹⁰⁶ See FED. R. EVID. 406; *id.* advisory committee’s note (quoting McCormick’s definition of habit as “the person’s regular practice of meeting a particular kind of situation with a specific type of conduct,” conduct whose doing “may become semi-automatic,” and adding that “[e]quivalent behavior on the part of a group is designated ‘routine practice of an organization’”).

From tort, we borrow its defining safeguard: recognition that is evidentiary, not dispositive—a functional norm can inform the standard of care, but it can never shield unreasonable or unlawful conduct. From the evidence rules, we borrow the proof path: routine-practice evidence that ties an agent’s regularities to the organization that deploys it. What neither supplies is the machinery for establishing that a practice binds—that machinery, built on belonging and knowing, is what Part III rebuilds.

C. Electronic Transactions Law: The Validity of Automated Acts

Contract and tort ask what an emergent practice means for the parties’ obligations. Electronic-transactions law and the law of agency ask two prior questions: whether an automated act counts at all, and whose act it is.

The first finds an answer in the Uniform Electronic Transactions Act (UETA), promulgated by the Uniform Law Commission in 1999 and enacted in every state but New York. The Act provides that a contract “may be formed by the interaction of electronic agents of the parties, even if no individual was aware of or reviewed” the agents’ actions. The federal E-SIGN Act of 2000, which backstops it for interstate commerce, adds that a contract’s validity may not be denied solely because an electronic agent participated in its formation, so long as the agent’s action “is legally attributable to the person to be bound.”¹⁰⁷ These frameworks settle the first question: automated acts are legally effective, no matter whose they turn out to be.¹⁰⁸ Nor is the answer stranded in

¹⁰⁷ See UNIF. ELEC. TRANSACTIONS ACT § 14(1) (UNIF. L. COMM’N 1999); *id.* § 14 cmt. 1 (the “requisite intention flows from the programing and use of the machine”). See also *id.* §§ 2(6), 9(a) (defining “electronic agent”); Electronic Signatures in Global and National Commerce Act, 15 U.S.C. §§ 7001(h), 7006(3) (2024). The drafters’ working premise, however, was the machine as a tool with “no independent volition”—the frame Part II.D’s attribution analysis must confront. See UNIF. ELEC. TRANSACTIONS ACT § 2 cmt. 5 (UNIF. L. COMM’N 1999); *id.* § 9 cmt. 1.

¹⁰⁸ UETA has been enacted in the other forty-nine states, the District of Columbia, Puerto Rico, and the U.S. Virgin Islands; New York alone retains its Electronic Signatures and Records Act (ESRA), which has no analogue to UETA’s automated-transaction or attribution provisions and whose qualification for E-SIGN’s reverse-preemption safe harbor is doubtful. See COMM. ON COM. L. & UNIF. STATE LAWS, N.Y.C. BAR ASS’N, SUPPORT FOR THE ENACTMENT OF THE UNIFORM ELECTRONIC TRANSACTIONS ACT IN NEW YORK 1 n.3, 6, 9 (Jan. 2026) (noting that “ESRA is silent” on attribution and that “this reverse preemption provision does not apply in New York”), https://www.nycbar.org/wp-content/uploads/2026/03/20221468-UETA_ESRA.pdf [<https://perma.cc/P465-DALF>]; N.Y. State Tech. Law §§ 301–309 (McKinney, Westlaw through L. 2026, ch. 184). E-SIGN thus supplies a federal floor for transactions in interstate commerce. See 15 U.S.C. §§ 7001(a), 7002(a) (2024).

1999: the drafters anticipated systems that would “act autonomously, and not just automatically,” and invited courts to “construe the definition of electronic agent accordingly.”¹⁰⁹

But this answer runs act by act: the statutes are calibrated to one record, one electronic agent, one principal. An emergent practice, by contrast, exists only in the aggregate, across records, agents, and counterparties—a standing rule that neither statute recognizes.

The second—whose act it is—the statutes decline to answer. Their attribution provisions are conduits, not answers: they route the question to existing law rather than resolve it. An electronic agent’s act is “attributable to a person if it was the act of the person,” a determination the drafters expressly left to “existing rules of law regarding attribution”; the comments’ closest approach to practice is their acknowledgment that “an established course of dealing between parties may result in a finding of attribution”—course of dealing, the very doctrine Part II.A surveys.¹¹⁰

From electronic-transactions law, we borrow the baseline: the settled validity of automated acts—for a practice can be recognized only if the acts composing it are legally effective at all. For attribution—and for the rules behind the acts—the law turns to the principles of agency.¹¹¹

D. Agency: Attributing Conduct to Principals

Agency law supplies the common law’s chief framework for attributing conduct to a legally responsible principal. Its threshold definition is built for persons—agency is a fiduciary relationship, and an AI system is not a person—but the three doctrines that matter here do not require a true agent.¹¹²

Apparent authority is the natural starting point. A principal can be bound when a third party reasonably believes an actor has authority and that belief is “traceable to the principal’s

¹⁰⁹See UNIF. ELEC. TRANSACTIONS ACT § 2 cmt. 5 (UNIF. L. COMM’N 1999).

¹¹⁰See UNIF. ELEC. TRANSACTIONS ACT § 9(a) & cmts. 1, 3 (UNIF. L. COMM’N 1999); 15 U.S.C. § 7001(h) (2024); *see also supra* note 85 (the Code’s course-of-dealing and course-of-performance doctrines and their ranking).

¹¹¹See *infra* Part III.B (developing the Attribution factor, for which these validity and attribution rules form the doctrinal baseline).

¹¹²See RESTATEMENT (THIRD) OF AGENCY §§ 1.01, 1.04(5) (AM. L. INST. 2006); *id.* §§ 2.03 & cmt. a, 2.05, 4.03 (apparent authority reaches “an agent or other actor”; estoppel operates precisely where no agency relationship exists; ratification reaches one who “acted or purported to act”).

manifestations.”¹¹³ Placing an employee in a position whose holders customarily have authority is such a manifestation; so is deploying an AI agent in a role. The doctrine already takes the content of reasonable belief from observed regularity—the move the recognition test generalizes.

Beyond apparent authority, two further doctrines are critical to attributing emergent AI behavior. First, *ratification*: a principal who, with knowledge of the material facts, affirms an agent’s prior act adopts that act as its own. Affirmance may be express, or it may lie in conduct treating the act as authorized or in retaining its benefits.¹¹⁴ A firm that observes a beneficial emergent practice—a novel routing protocol, say—and knowingly continues to profit from it invites that analysis: awareness of “ongoing conduct encompassing numerous acts,” coupled with a failure to terminate, “may constitute ratification.”¹¹⁵ Second, *estoppel*: even a principal who has made no manifestation of authority at all may be barred from denying it, if a third party detrimentally relied on a belief the principal intentionally or carelessly caused, or, “having notice of such belief,” failed to take reasonable steps to correct.¹¹⁶ Apparent authority, ratification, and estoppel thus supply three routes for linking an AI agent’s conduct to a responsible principal—the doctrinal home of the F-NORM test’s Attribution factor.¹¹⁷

Yet agency, too, is calibrated to a human tempo and a human unit of analysis. Ratification’s knowledge and estoppel’s notice both presuppose a principal with time to learn what its agent has done; a practice that forms and dissolves within an interaction window may never cross a human desk. All three doctrines attach to a discrete *act*—this act, by this actor, for this principal. None reaches a standing *practice*, still less one that emerges across many firms’ agents, where there is no single principal to charge.¹¹⁸

¹¹³ See RESTATEMENT (THIRD) OF AGENCY § 2.03 (AM. L. INST. 2006); *id.* § 1.03 & cmt. b (defining “manifestation” to include conduct, and explaining that “placing the agent in such a position”—one “in which holders of the position customarily have authority”—“constitutes a manifestation”).

¹¹⁴ See RESTATEMENT (THIRD) OF AGENCY § 4.01 (AM. L. INST. 2006); *id.* § 4.06 (ratification is effective only with knowledge of material facts).

¹¹⁵ See RESTATEMENT (THIRD) OF AGENCY § 4.01 cmt. d (AM. L. INST. 2006) (adding that conduct “that can be otherwise explained may not effect ratification”).

¹¹⁶ See RESTATEMENT (THIRD) OF AGENCY § 2.05 (AM. L. INST. 2006).

¹¹⁷ See *infra* Part III.B.

¹¹⁸ See RESTATEMENT (THIRD) OF AGENCY §§ 4.01(1), 4.07 (AM. L. INST. 2006) (ratification operates on discrete acts,

From agency, therefore, we borrow the vocabulary of attribution—manifestation, ratification, reliance—and supply what these doctrines assume: a way to establish what the practice was while it lasted, and who can fairly be charged with it.

E. The Outer Boundaries: Mandatory Public Law

The preceding sections ask what the recognition of an emergent practice can do for a court—fix the meaning of an agreement, fill its gaps, inform the standard of care, validate an automated act, and connect it to a responsible principal. None asks whether the practice itself is one the law will tolerate. For practices that emerge among AI agents, that question is acute: the qualities that make a practice a candidate for recognition—regularity, reliable observance, responsiveness to what other agents do—are also the qualities that make a cartel work. The answer comes from public law—law that private ordering cannot displace—and it runs along three lines: how a practice arises, what it does to those it reaches, and what an agent is in law.

Antitrust law draws the first line. Its core is settled: “the machinery employed by a combination for price-fixing is immaterial.”¹¹⁹ An algorithm is such machinery. Courts have accordingly allowed hub-and-spoke theories to proceed where the alleged hub is a pricing tool rather than a person.¹²⁰ An emergent practice that amounts to price-fixing is unlawful however organically it arises, and recognition would be no shelter. Thus, recognition must be conditioned on legality, and the condition must reach every form of coordination the antitrust laws condemn, not price-fixing

taken in their entirety).

¹¹⁹ See *Socony-Vacuum*, 310 U.S. 150, 223; *Apple*, 791 F.3d 290, 314, 322 (affirming liability in a hub-and-spoke price-fixing conspiracy).

¹²⁰ See *Cornish-Adebiyi v. Caesars Entm’t, Inc.*, No. 24-3006, 2026 WL 2182291, at *12 (3d Cir. July 29, 2026) (reversing dismissal of § 1 claims where casino-hotels allegedly pooled real-time nonpublic pricing and occupancy data through a shared pricing algorithm: “plaintiffs have surely raised a plausible inference of collusion under Section 1 of the Sherman Act even though the alleged hub is named ‘Rainmaker’ rather than ‘Bob’ ”); *In re RealPage, Inc., Rental Software Antitrust Litig.* (No. II), 709 F. Supp. 3d 478, 501–03 (M.D. Tenn. 2023) (holding an algorithmic hub-and-spoke conspiracy plausibly alleged); Statement of Interest of the United States, *supra* note 13, at 2 (“Automating an anticompetitive scheme does not make it less anticompetitive.”). The frontier is pleading-sensitive rather than settled: where a complaint alleges only common vendor use, without pooled nonpublic data, the theory has failed. See cases cited *supra* note 21.

alone.¹²¹ It must also operate as a threshold rather than a factor: a cartel’s dealings can be foreseeable, regular, attributable, material, and verifiable—every criterion the test weighs—so a practice that fails on legality cannot be saved by how well it does on the rest. That is the work the Legality Screen does in the F-NORM test. But nothing in the test makes a practice unlawful; the antitrust laws do that.

Civil-rights and consumer-protection statutes draw the second line. An emergent practice can be collusion-free, innocent in origin, and successful at its task—and still unlawful, because liability here attaches to effects. A practice that produces discriminatory outcomes may be unlawful under the civil-rights regimes that reach disparate impact, without any showing of intent, and its functional effectiveness is not itself a defense.¹²² And the duties are non-waivable: no express agreement can displace what the Fair Housing Act, the Equal Credit Opportunity Act, and Title VII of the Civil Rights Act of 1964 require—still less an informal practice.¹²³ The Federal Trade Commission Act extends the same protection to consumers at large, reaching practices that are unfair or deceptive whether or not anyone set out to make them so.¹²⁴ Two things follow for the doctrine. The Legality Screen must reach effects-based violations no less than unlawful coordination—a screen that policed formation alone would clear practices these statutes condemn. And the ordering the doctrine adopts—practice ordinarily yielding to express terms, both always yielding to mandatory law—is inherited rather than invented: contract law supplied the first half, and these statutes supply the second.

¹²¹ See *United States v. Container Corp. of Am.*, 393 U.S. 333, 335–38 (1969) (condemning information exchanges that stabilize prices); *In re Text Messaging Antitrust Litig.*, 782 F.3d 867, 879 (7th Cir. 2015) (“Tacit collusion, also known as conscious parallelism, does not violate section 1 of the Sherman Act. Collusion is illegal only when based on agreement.”); *id.* at 871 (defining conscious parallelism as “coordinating their pricing without an actual agreement to do so”); *Apple*, 791 F.3d at 315 (quoting *Apex Oil Co. v. DiMauro*, 822 F.2d 246, 253 (2d Cir. 1987)) (requiring “additional circumstances, often referred to as ‘plus’ factors,” before an agreement may be inferred from parallel conduct).

¹²² See 42 U.S.C. § 2000e-2(k) (2024) (codifying disparate-impact liability under Title VII); *Tex. Dep’t of Hous. & Cmty. Affs. v. Inclusive Communities Project, Inc.*, 576 U.S. 519, 539–40 (2015) (recognizing disparate-impact claims under the Fair Housing Act).

¹²³ See Fair Housing Act § 804, 42 U.S.C. § 3604 (2024); Equal Credit Opportunity Act § 701, 15 U.S.C. § 1691 (2024); Civil Rights Act of 1964 § 703, 42 U.S.C. § 2000e-2 (2024); see also *Alexander v. Gardner-Denver Co.*, 415 U.S. 36, 51–52 (1974) (holding that there can be no prospective waiver of an employee’s rights under Title VII).

¹²⁴ See 15 U.S.C. § 45(a)(1), (n) (2024).

Intellectual property law draws the third line, and it is a line of a different kind. It bars no practice. Instead, it forecloses an implication a doctrine of recognition might be taken to carry: that recognition could open the door to legal personhood for AI agents. Copyright protection requires human authorship.¹²⁵ Patent law requires that an “inventor” be a natural person.¹²⁶ These rules about *status* rather than conduct fix what an agent is and is not, irrespective of the proposed doctrine. They mark a boundary: recognizing a practice can attach consequences to conduct, but it cannot confer authorship, inventorship, or personhood on the agents whose interactions produced it.

From public law, then, we borrow boundaries; these rules need no recalibration because they never rested on the recognition of emergent practices. None asks who belongs, who knew, or how long a practice held—the questions on which existing doctrines strain. The proposed doctrine widens what a court may recognize; it does not widen what the law permits.

In sum, existing law already possesses the raw materials: awareness and notice, regularity of observance, the primacy of express terms, the commercial significance of a practice, evidentiary weight that is probative but not dispositive, a proof path for routine practice, validity for automated acts, routes of attribution, and public-law boundaries. What it lacks is a single frame: the doctrines that recognize practices presuppose human communities, and the doctrines that attribute conduct reach only discrete acts—while a practice that emerges across many firms’ agents answers to no single principal at all. Part III takes these materials and reassembles them into a standard tailored to the frictions identified above: *transience*, *opacity*, and *nonhuman origin*.

¹²⁵ See *Burrow-Giles Lithographic Co. v. Sarony*, 111 U.S. 53, 58 (1884) (grounding authorship in “original intellectual conceptions of the author”); *Thaler v. Perlmutter*, 130 F.4th 1039, 1041 (D.C. Cir. 2025) (affirming the human-authorship requirement), *cert. denied*, No. 25-449 (U.S. Mar. 2, 2026); Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence, 88 Fed. Reg. 16,190, 16,191–92 (Mar. 16, 2023).

¹²⁶ See *Thaler v. Vidal*, 43 F.4th 1207, 1210, 1213 (Fed. Cir. 2022) (holding that “Congress has determined that only a natural person can be an inventor, so AI cannot be”).

III. THE DOCTRINE OF FUNCTIONAL NORMS

The last Part has shown that to treat emergent practices as legally operative, the law does not need new *values*, but it does need a recalibrated *test*. A usage of trade accretes across repeated dealings; an emergent practice stabilizes in minutes. A custom is visible to the naked eye; an algorithmic routine surfaces only to a forensic audit, and its reasons not at all. This Part articulates that test: the doctrine of functional norms.

A. Formal Definition

We define a functional norm and specify the legal effect of recognizing one as follows:

A *functional norm* is an emergent practice that, on balance, is (1) *Foreseeable* to affected parties, (2) *Regular* within the relevant interaction window, (3) *Attributable* to a legally responsible principal, (4) *Material* to outcomes, and (5) *Verifiable* through reliable records. If such a practice is also (6) *Consistent* with express contractual terms and (7) cleared by a *Legality Screen* that bars recognition of anticompetitive or otherwise unlawful conduct, a court may treat it as legally operative for purposes of interpretation and gap-filling, and as probative (but not dispositive) evidence of the standard of care.

B. The Recognition Test: The Seven F-NORM Criteria

Embedded in that definition is the operational heart of the doctrine—the F-NORM test, a standard for determining when an emergent practice should be granted legal effect. The five weighted factors assess whether a practice in fact functions as a norm in the parties’ interaction, and the two threshold “gates” screen for consistency with private ordering and public law. Each criterion tracks a familiar legal value.¹²⁷ Foreseeability protects parties from unfair surprise (notice). Regularity

¹²⁷For the doctrinal foundations of these values as developed in Parts I and II: on reliance and regularity, party autonomy, and unfair surprise, see U.C.C. § 1-303(c), (e), (g) (AM. L. INST. & UNIF. L. COMM’N 2022); on accountability through

captures reliance interests. Attribution enforces accountability by tying behavior to a responsible principal. Materiality measures significance. Verifiability ensures administrability. The gates respect party autonomy (contract primacy) and the supremacy of mandatory public law.

The analysis proceeds sequentially. The gates come first and are dispositive: if the practice fails either the Consistency gate or the Legality Screen, the inquiry ends. Each is a limit that the law already places on what a practice can do, and carries the tilt the law gives it. If the practice passes both gates, the court then undertakes a holistic analysis of the five weighted factors. The proponent bears the burden of establishing the facts underlying the weighted factors by a preponderance of the evidence and of showing that, on balance, the practice functions as a norm.¹²⁸ The strength of the showing then calibrates the scope of the legal effect granted, as Part III.C elaborates.

1. *Foreseeability*:

Foreseeability asks whether a reasonable party, in context, could have anticipated the practice's emergence. The factor carries into algorithmic settings the notice and foreseeability principles that anchor fairness in contract and tort law, so that no hidden or arbitrary practice ambushes a party that had no opportunity to account for it.¹²⁹

The proponent proves foreseeability with sources available to a counterparty—for example, API documentation describing expected agent behaviors, platform-wide bulletins announcing new operational parameters, the known defaults of widely used

attribution, *see* RESTATEMENT (THIRD) OF AGENCY § 2.03 (AM. L. INST. 2006); on materiality, *cf.* *Nanakuli*, 664 F.2d 772, 780; on administrability, *see* FED. R. EVID. 901(b)(9), 902(13)–(14); on the supremacy of public law, *see supra* notes 119–124 and accompanying text.

¹²⁸ *See* U.C.C. § 1-303(c) (AM. L. INST. & UNIF. L. COMM'N 2022) (“The existence and scope of such a usage must be proved as facts.”); RESTATEMENT (SECOND) OF CONTRACTS § 222(2) (AM. L. INST. 1981) (the existence and scope of a usage of trade “are to be determined as questions of fact”).

¹²⁹ *See Nanakuli*, 664 F.2d at 806 (Kennedy, J., concurring specially) (conditioning practice-based terms on “well-established custom and usage or other objective standards of which the parties had clear notice”); *cf.* U.C.C. § 1-303(g) (AM. L. INST. & UNIF. L. COMM'N 2022) (requiring notice to prevent unfair surprise when offering evidence of usage of trade); RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 3 (AM. L. INST. 2010) (making “the foreseeable likelihood that the person’s conduct will result in harm” a primary factor in the negligence assessment).

software libraries, a prior course of dealing, or patterns observable in public telemetry. The counterparty can rebut the showing with evidence that the practice was generated by hidden configurations, nonpublic A/B testing, or other sources not reasonably discoverable.

2. *Regularity Within the Interaction Window:*

Regularity measures how consistently the practice repeated *within* the relevant interaction window rather than across months or years. The recalibration adapts custom to practices that are *transient*: critically important for a brief period, then gone once the underlying coordination problem is resolved. The proponent selects the window and must justify the selection, which is open to challenge—as too narrow, too convenient, or unrepresentative of the parties’ interaction.¹³⁰

The primary data detailed in Part I.C can prove regularity directly: logs showing a stable pattern, low variance in agent response times, and, most powerfully, anticipatory behavior from counterparties who act as if they expect the pattern to continue. Rebuttal comes from evidence that the pattern was a one-off incident, was not statistically distinguishable from ordinary variation, or was merely an artifact of an external constraint such as network throttling rather than a genuine interactional practice.

3. *Attribution:*

Attribution asks whether the practice traces to the behavior of agents for whom a specific, legally responsible principal must answer.¹³¹ That trace secures accountability: it ties algorithmic behavior to a person or entity and prevents an agent from

¹³⁰ See *supra* note 25 (defining the interaction window as the bounded period of interaction giving rise to the alleged norm); see also RESTATEMENT (SECOND) OF CONTRACTS § 222(1) (AM. L. INST. 1981) (providing that a usage of trade “may include a system of rules regularly observed even though particular rules are changed from time to time”).

¹³¹ Attribution does not require proof that the principal explicitly programmed the specific emergent behavior. See *supra* notes 113–116 and accompanying text (deployment in a role supplying the manifestation for apparent authority, with knowing continued deployment and retention of benefits supporting attribution by ratification or estoppel).

being treated as legally untethered. Attribution builds directly on the principles of electronic-transactions and agency law discussed in Parts II.C and II.D, and answers the multi-principal problem the latter poses.¹³² Deploying an agent in a role is itself a manifestation, so the factor does not require that anyone watched; knowledge of the practice strengthens the showing rather than conditioning it. Where a practice emerges across many firms' agents, the factor runs principal by principal: the question is not whether a single principal owns the practice, but whether a given principal's agents participated in it regularly enough to be charged with it.

The secondary data detailed in Part I.C can prove attribution: version-control histories, deployment manifests, and signed configuration bundles link an agent's behavior to a specific principal's settings or accepted defaults. Routine-practice evidence supplies the courtroom vehicle, tying an agent's regularities to the organization that deploys it.¹³³ Evidence that the behavior did not originate with the principal's agents rebuts the showing—for example, that the practice was an artifact of a third-party platform's hidden traffic-shaping policies.

4. *Materiality*:

Materiality measures effect: whether the practice shaped the outcome of the interaction in more than a de minimis way. Regularity alone will not do; the effect must be concrete—on price, resource allocation, timing, or risk. This ensures that courts give legal weight only to practices that are truly “functional.”

The proponent proves materiality with causal tools: A/B tests that isolate the effect, counterfactual simulations, or event studies that measure the change in outcomes when the practice emerged or was disrupted. The same tools can rebut that showing

¹³² See RESTATEMENT (THIRD) OF AGENCY §§ 1.01–1.03, 2.03, 2.05, 4.01 (AM. L. INST. 2006) (supplying the attribution pathways: manifestation, apparent authority, estoppel, and ratification); see also *supra* note 26 (electronic-transactions authority routing the attribution of automated acts to existing law).

¹³³ See *supra* note 106 and accompanying text.

by revealing a de minimis effect or outcomes better explained by other confounding factors.

5. *Verifiability*:

Verifiability asks whether a court or a neutral expert can reconstruct the claimed practice from reliable, contemporaneous records and confirm its existence. That capability answers the “black-box” problem and ensures that recognition is based on objective, testable evidence, not speculation.¹³⁴

Verifiability depends on the evidentiary substrate detailed in Part I.C. The proponent must be able to produce records supporting its claim—tamper-evident logs, hashed configuration files, and auditable telemetry. The risk of inadequate records falls on the proponent: a practice that cannot be reconstructed has not been proved. Rule 37(e) may supply remedies where the deficiency results from the opposing party’s failure to preserve, but only for records lost after a duty to preserve attached.¹³⁵

6. *Consistency with Express Terms (The First Gate)*:

Consistency bars recognition of an emergent practice that directly contradicts a clear, express term of a governing agreement, and so respects the primacy of private ordering. A court resolves the consistency question by construction rather than by weight: express terms and applicable practices are construed as consistent whenever reasonable, and the express terms prevail only where that reading is unreasonable.¹³⁶

The construction therefore tilts toward the practice, and where the evidence permits

¹³⁴ See *Daubert*, 509 U.S. 579, 590, 593 (requiring that expert evidence rest on “more than subjective belief or unsupported speculation” and be testable).

¹³⁵ See FED. R. CIV. P. 37(e); see also *supra* note 71 (discussing preservation duties and spoliation sanctions).

¹³⁶ See U.C.C. § 1-303(e) (AM. L. INST. & UNIF. L. COMM’N 2022) (requiring that express terms and applicable practice sources “be construed whenever reasonable as consistent with each other,” with express terms prevailing only “[i]f such a construction is unreasonable”); *id.* § 1-303(f) (making a course of performance, subject to § 2-209, relevant to show a waiver or modification of an inconsistent term).

more than one reasonable construction, the question survives judgment as a matter of law.¹³⁷

An emergent practice can give particular meaning to the parties' terms, qualify them, or fill a gap where the agreement is silent. The gate closes only on direct conflict with a clear, express term—for example, a protocol allocating risk in one way while the agreement expressly allocates it in another. Even then, the same practice may still show a waiver or modification of that term under ordinary principles, which are available for any inconsistent term and so do not track the gate.

7. *Legality Screen (The Second Gate)*:

The *Legality Screen* is the doctrine's most important safeguard. The practice must neither violate nor facilitate the violation of mandatory public law.

Part II.E supplies the content of that law: a court must deny recognition to a practice that amounts to, or facilitates, illegal price-fixing, unlawful discrimination, or a violation of consumer protection law, regardless of how well it satisfies the other criteria.¹³⁸ The gate closes only on an established violation; a practice that is merely suspect does not fail it.¹³⁹ Legality is policed by the court rather than allocated to a party: either side may raise it, and a court may raise it on its own, because the interest is not the parties' to waive.

Part I.D details how each dimension of a practice is established from the record and which criteria it informs.

¹³⁷ See *supra* note 90 and accompanying text (reinstating a jury finding that a trade usage could reasonably be construed as consistent with an express price term).

¹³⁸ See *supra* notes 119–124 and accompanying text (collecting the antitrust, civil-rights, and consumer-protection backstops that bound recognition); cf. U.C.C. § 1-303 cmt. 5 (AM. L. INST. & UNIF. L. COMM'N 2022) (noting that “the anciently established policing of usage by the courts is continued” for cases where “an unconscionable or dishonest practice should become standard”); *id.* § 1-303 cmt. 3 (distinguishing usage from “mandatory rules of law . . . whose very office is to control and restrict the actions of the parties”).

¹³⁹ Cf. RESTATEMENT (SECOND) OF CONTRACTS § 178(1) (AM. L. INST. 1981) (making a contract term unenforceable on grounds of public policy where legislation so provides or the interest in its enforcement “is clearly outweighed in the circumstances by a public policy against the enforcement of such terms”).

C. Recognition as Calibrated and Context-Specific

Recognition is not necessarily all-or-nothing. What varies is scope, not the showing required: a practice that does not, on balance, function as a norm is not recognized at any tier. Where the weighted factors are mixed or where public-policy concerns short of illegality counsel restraint, a court may grant *graduated recognition*—acknowledging a practice’s existence for limited purposes while withholding broader legal effects. The range runs from *narrow recognition*, which credits the practice’s existence and lets it be weighed without giving it operative effect, to *full recognition*, which treats the practice as legally operative. This calibrated approach has deep roots. In tort law, custom receives this treatment: once a practice’s existence is credited, it becomes probative of reasonable care without ever becoming the measure of it.¹⁴⁰ And in contract law, usage may explain or supplement a written agreement, yet it yields to express terms with which it cannot reasonably be construed as consistent.¹⁴¹ The doctrine takes the same flexibility, fitting legal consequences to the strength of the evidence and the context of the dispute.

Recognition also neither creates a new substantive right nor confers anything on the agents themselves: a functional norm makes an agent’s acts matter without making the agent an author, an inventor, or a person. The doctrine is designed to help courts adjudicate disputes arising from AI-mediated interactions.¹⁴² For instance:

- *In Contract Law*, a recognized functional norm can give particular meaning to the parties’ terms or supplement the agreement where it is silent, consistent with the U.C.C. hierarchy.

Where the practice is bilateral, it can also establish a course of performance, which may

¹⁴⁰ See *Trimarco*, 436 N.E.2d 502, 505–06 (holding that “once its existence is credited, a common practice or usage is still not necessarily a conclusive or even a compelling test of negligence”); *The T.J. Hooper*, 60 F.2d 737, 740 (holding that a calling “never may set its own tests, however persuasive be its usages”).

¹⁴¹ See U.C.C. § 1-303(d)–(e) (AM. L. INST. & UNIF. L. COMM’N 2022) (subsection (d) permitting practice to give particular meaning to and supplement or qualify agreement terms; subsection (e) giving express terms priority where a consistent construction is unreasonable); *Columbia Nitrogen*, 451 F.2d 3, 9–10 (holding that “a finding of ambiguity is not necessary for the admission of extrinsic evidence about the usage of the trade and the parties’ course of dealing,” while requiring that such evidence “reasonably can be construed as consistent with the express terms of the agreement”).

¹⁴² See *infra* Part VI (answering the democratic-legitimacy objection that the doctrine would allow machines to make law).

show a waiver or modification of a term.¹⁴³

- *In Tort Law*, a recognized functional norm can serve as *probative but not dispositive evidence* of the standard of care.¹⁴⁴ Evidence that a safety-enhancing practice prevailed across the relevant interaction window, for example, can help a factfinder determine what a reasonable operator in the defendant's position would have done. A court retains the authority, consistent with *The T.J. Hooper*, to find that the norm itself was negligent.¹⁴⁵ Recognition is evidence of the standard; it is not a safe harbor.

IV. THE DOCTRINE IN ACTION: FOUR CASE STUDIES

This Part moves from theory to practice through four case studies: a practice the court recognizes outright, a course of performance it recognizes to interpret express terms but not to override them, an ambiguous practice it recognizes only narrowly, and a collusive practice that mandatory public law bars it from recognizing at all. The studies are stylized, designed to show how a court would apply the F-NORM framework. Each assumes that a dispute has arisen and that discovery has produced the relevant records. Each follows the staged approach: the two gates are questions the court resolves at summary judgment, and if a claimed norm passes both, the court asks whether, on balance, the practice functions as a norm.¹⁴⁶

¹⁴³See *supra* Part I.B (defining bilateral practices as those shaped by the two parties themselves, arising as a course of performance within a single agreement); *supra* note 87 and accompanying text (giving effect to a course-of-performance waiver notwithstanding a no-oral-modification clause); *supra* Part III.B (the waiver route being available for any inconsistent term, and so not tracking the Consistency gate); *infra* Part IV.B (applying these principles in the cloud-SLA case study).

¹⁴⁴See *infra* Part IV.A (applying this standard in the drone-collision case study).

¹⁴⁵See *The T.J. Hooper*, 60 F.2d at 740; RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 13 cmt. b (AM. L. INST. 2010) (explaining that compliance with custom does not conclusively establish non-negligence because “[p]ossibly, the entire community or industry has lagged”).

¹⁴⁶See *infra* Part V.A (proposing this staged adjudication).

A. Case Study 1: Benign Coordination

The first case study returns to the setting of the opening example and presents the clearest case for recognition: a practice that emerges to solve a coordination problem. Two competing logistics companies, “FleetA” and “FleetB,” operate autonomous delivery drone fleets in a dense urban area.¹⁴⁷ Over several weeks, their fleet management systems converge on an emergent “altitude-for-direction” practice in a key corridor of uncontrolled airspace that only their two fleets regularly use: northbound drones fly at 200 feet AGL, southbound at 400.¹⁴⁸ No contract or regulation specifies how drones transiting in opposite directions are to separate.¹⁴⁹ Yet the practice works: once it takes hold, both systems’ incident logs record markedly fewer near misses.¹⁵⁰

More than a month into the practice’s operation, a FleetA drone flies northbound at 400 feet AGL and collides with a southbound FleetB drone.¹⁵¹ FleetB sues FleetA for negligence, seeking

¹⁴⁷The vignette assumes the fleets hold the authorizations that routine beyond-visual-line-of-sight operation requires, *cf.* 14 C.F.R. § 107.31 (2025) (visual-line-of-sight requirement), and that the FAA’s pending framework for such operations has not yet prescribed a corridor deconfliction protocol. *See* Normalizing Unmanned Aircraft Systems Beyond Visual Line of Sight Operations, 90 Fed. Reg. 38,212 (Aug. 7, 2025) (proposed rule); *see also* 91 Fed. Reg. 3,695 (Jan. 28, 2026) (reopening comment period).

¹⁴⁸This “altitude-for-direction” practice is a direct analogue to manned-flight VFR directional altitude rules and is consistent with the small-UAS ceiling. *See* 14 C.F.R. § 91.159 (2025) (establishing directional altitudes for VFR flight above 3,000 feet AGL); *id.* § 107.51(b) (general 400-ft AGL limit). The FAA anticipates operators settling such practices among themselves: in the corridor-based operations it envisions for cooperative low-altitude environments, of which urban air mobility is the most fully specified, the collaborative development of Cooperative Operating Practices “would allow for stakeholders to agree on norms of interactions, which may reduce the need for ATC tactical control of individual flights and management of access.” *See* FED. AVIATION ADMIN., URBAN AIR MOBILITY (UAM) CONCEPT OF OPERATIONS 9 (Version 2.0, Apr. 26, 2023), <https://www.faa.gov/sites/faa.gov/files/Urban-Air-Mobility-Concept-of-Operations-2.0.pdf> [<https://perma.cc/64HR-PQ3X>].

¹⁴⁹Federal regulation imposes only a general duty to yield and to avoid creating a collision hazard. *See* 14 C.F.R. § 107.37 (2025) (requiring each small unmanned aircraft to yield the right of way to all aircraft and barring operation “so close to another aircraft as to create a collision hazard”). Because § 107.37 fixes the duty without specifying its content, it leaves open what mechanism of separation (distance, time, elevation) reasonable care would require—the question the emergent practice answers.

¹⁵⁰The evidentiary substrate is made practical by the FAA’s Remote ID rule, which requires public broadcast of message elements (including UA ID, position, altitude, velocity, and timestamp). *See* Remote Identification of Unmanned Aircraft, 86 Fed. Reg. 4,390, 4,507–08 (Jan. 15, 2021) (codified at 14 C.F.R. §§ 89.305, 89.310(g)(1), 89.315).

¹⁵¹The collision risk is not theoretical. In 2017, a civilian DJI Phantom drone collided in mid-air with a U.S. Army Black Hawk helicopter near Hoffman Island, New York. *See* NAT’L TRANSP. SAFETY BD., AVIATION INVESTIGATION FINAL REPORT (No. DCA17IA202, Dec. 14, 2017), <https://data.nts.gov/carol-repgen/api/Aviation/ReportMain/GenerateNewestReport/96058/pdf> [<https://perma.cc/W76M-8R2X>]. And the structural fix that the practice arrived at ad hoc is one that safety authorities often reach for deliberately. After a January 2025 mid-air collision near Ronald Reagan Washington National Airport, the NTSB urgently recommended segregating airplane and helicopter traffic by route and timing, finding existing vertical separation insufficient. *See* NAT’L TRANSP. SAFETY BD., DECONFLICT

damages for its destroyed drone and cargo, and asks the court to recognize the practice as a functional norm and treat it as evidence of the standard of care.

FleetB, as proponent, defines the interaction window as the thirty days preceding the collision—a period that opens only after the practice has stabilized.¹⁵² The practice clears the Consistency gate: the two fleets have no agreement with one another whose express terms it could contradict, and neither company’s customer agreements purport to allocate corridor altitudes.¹⁵³ It also clears the Legality Screen: the practice violates no mandatory public law, as both altitudes fall within the small-UAS ceiling and no federal aviation regulation prescribes an altitude-allocation protocol for the corridor.¹⁵⁴

The weighted factors strongly support recognition:

- *Foreseeability* is high: each fleet’s own systems followed the altitude-for-direction practice for weeks and logged it, and every drone broadcast its position continuously over public Remote ID, leaving the pattern observable to any operator monitoring the corridor.¹⁵⁵
- *Regularity* is also high: an audit of the parties’ cryptographically time-stamped and hashed flight logs shows that over 99 percent of corridor flights adhered to the practice throughout the window.
- *Attribution* is clear: each fleet’s flight paths trace through the fleet management software to the company that configured and deployed it, as evidenced by the company’s deployment manifests and configuration histories.
- *Materiality* is compelling: an event study spanning the weeks before and after stabilization

AIRPLANE AND HELICOPTER TRAFFIC IN THE VICINITY OF RONALD REAGAN WASHINGTON NATIONAL AIRPORT 9 (Aviation Investigation Report AIR-25-01, Mar. 7, 2025), <https://www.nts.gov/investigations/AccidentReports/Reports/AIR2501.pdf> [<https://perma.cc/NJG7-BGMX>] (Safety Recommendations A-25-1 and A-25-2).

¹⁵²On the interaction-window concept, *see supra* note 25. On window selection and its contestability, *see supra* Part III.B. Evidence bearing on Materiality may reach back before the window, since the practice’s effect can be measured only against a pre-stabilization baseline.

¹⁵³Where no express term is in play, the gate is satisfied without more. *See supra* Part III.B.

¹⁵⁴*See* 14 C.F.R. §§ 107.37, 107.51(b) (2025).

¹⁵⁵Remote ID altitude need be accurate only to within 150 feet of true geometric altitude, with 95 percent probability. *See* 14 C.F.R. § 89.310(h)(3) (2025). No single broadcast cleanly resolves even a 200-foot split, but the pattern does: the mean altitude conditional on heading is estimable to far better precision than any one observation.

shows a statistically significant reduction in near-miss alerts—from an average of about four alerts per day beforehand to about 0.3 afterward.

- *Verifiability* is straightforward: the same logs can be independently authenticated and the corridor’s traffic reproduced in a controlled replay from the parties’ preserved data.¹⁵⁶

On balance, the practice functions as a norm; the court grants full recognition and, in FleetB’s negligence action, treats it as probative evidence of the standard of care. That only two operators share the corridor—a micro-practice in Part I.B’s taxonomy—is no obstacle: the interaction window, not a trade or locality, fixes the relevant community, and § 13 asks about the custom “of others in like circumstances,” not industry-wide adoption.¹⁵⁷ FleetA’s deviation, traced to the previous night’s configuration push, is strong evidence of breach—but the norm is a benchmark for the factfinder, not a verdict.

The outcome would be different, however, if an FAA directive had explicitly prohibited such altitude differentiation. Then, the practice would fail the Legality Screen, as an express regulatory command prevails regardless of the practice’s safety benefits. The consequences would not stop at lost evidentiary support. FleetB’s own adherence to the prohibited practice would be the unexcused violation, and FleetA could not be negligent for departing from an unlawful norm.¹⁵⁸

B. Case Study 2: A Course of Performance Versus Express Terms

The second case study moves from tort to contract. Two sophisticated firms—“Orchestrio,” a cloud platform, and “FinCo,” a quantitative trading firm—have a service-level agreement (SLA) for high-priority compute.¹⁵⁹ The SLA guarantees Gold-tier jobs priority over the lower tiers. Within

¹⁵⁶Remote ID requires broadcast, not retention; the preserved logs are the operators’ own, maintained under internal retention policies and, after the incident, litigation holds. On preservation, *see infra* Part V.A. On timestamp sealing and log integrity, *see supra* note 70.

¹⁵⁷*See* RESTATEMENT (THIRD) OF TORTS: LIAB. FOR PHYSICAL & EMOTIONAL HARM § 13(b) & cmts. c–d (AM. L. INST. 2010) (observing that “rules of the road—such as driving on the right—can develop on private roadways ungoverned by public regulations”); *see also supra* Part I.B (distinguishing micro-, meso-, and macro-practices); *supra* Part III.C.

¹⁵⁸*See supra* note 103 (custom cannot excuse violation of a statutory or regulatory duty).

¹⁵⁹Cloud computing Service Level Agreements (SLAs) often guarantee high-level performance metrics while leaving granular implementation details, such as scheduler behavior, unspecified. *See, e.g.,* AMAZON WEB SERVS., *Amazon*

the Gold tier, scheduling is “first-come, first-served (FCFS) where feasible.” The SLA contains a merger clause and no clause addressing the parties’ course of performance.

After deployment, Orchestrio’s AI scheduler converges on an emergent “age-based fair-share” practice during periods of high congestion.¹⁶⁰ Rather than letting a single large job block the queue, the system rotates small time-slices among all concurrent Gold-tier tenants. Over ninety trading days, both parties’ agents settle into this rhythm. FinCo’s systems learn to break large jobs into micro-batches timed to the scheduler’s cadence, and queue logs show that during peak load, Gold-tier jobs alternate in a stable, predictable pattern.

On a volatile market morning, demand surges across the Gold tier and the practice allocates time-slices among the competing tenants. FinCo places a critical risk-control job that completes several seconds later than it would have under strict sequencing—a delay that is trivial in absolute terms but material to the firm whose exposure limits are enforced by that job. FinCo sues for breach of contract, arguing the SLA guarantees strict FCFS ordering. Orchestrio counters that the “where feasible” language is ambiguous during congestion, and that the practice—established through a clear course of performance—is a functional norm that gives the term meaning.¹⁶¹

Orchestrio, as proponent, defines the interaction window as the ninety trading days preceding the disputed delay. The practice clears the Legality Screen: no mandatory public-law rule bars a platform from time-slicing compute within a service tier, and the practice neither is nor functions

Compute Service Level Agreement (last updated May 25, 2022), <https://aws.amazon.com/compute/sla> [<https://perma.cc/SX7H-BXH6>] (promising availability and credits but silent on intra-tier scheduling); GOOGLE CLOUD, *Compute Engine Service Level Agreement* (last modified Mar. 4, 2025), <https://cloud.google.com/compute/sla> [<https://perma.cc/BF5X-EH26>] (similar).

¹⁶⁰The “age-based fair-share” practice is analogous to real-world schedulers designed to prevent tenant starvation and ensure predictable performance. See LINUX KERNEL DOCUMENTATION, *CFS Scheduler* (last visited Dec. 20, 2025), <https://www.kernel.org/doc/html/latest/scheduler/sched-design-CFS.html> [<https://perma.cc/TY5C-T3CF>]; APACHE HADOOP, *Fair Scheduler* (last visited Dec. 20, 2025), <https://hadoop.apache.org/docs/stable/hadoop-yarn/hadoop-yarn-site/FairScheduler.html> [<https://perma.cc/DY79-C757>] (sharing resources so applications receive equal shares over time while not starving long-lived apps); SCHEDMD, *Multifactor Priority Plugin* (last visited Dec. 20, 2025), https://slurm.schedmd.com/priority_multifactor.html [<https://perma.cc/GR3U-HFP9>] (increasing priority with queue wait time/“age”).

¹⁶¹The legal analysis draws on the U.C.C.’s framework for using course of performance to interpret terms and on its common-law analogue: the SLA is a services contract, so Article 2 supplies the framework by analogy, with the Restatement furnishing the parallel rules of decision. See U.C.C. § 1-303 (AM. L. INST. & UNIF. L. COMM’N 2022); *id.* § 2-202 cmt. 2; RESTATEMENT (SECOND) OF CONTRACTS §§ 202(4), 203(b), 219–223 (AM. L. INST. 1981); *Nanakuli*, 664 F.2d 772, 780, 794–96; *Columbia Nitrogen*, 451 F.2d 3, 8–10; *supra* Part II.A.

as a horizontal restraint among competitors—the feature that condemns the practice in Case Study 4.¹⁶² The core question is Consistency—contradiction or interpretation. The governing standard asks whether the express term and the course of performance can “be construed whenever reasonable as consistent with each other,” and the limit courts have drawn is that a practice may qualify an express term so long as it does not “totally negate” it.¹⁶³ The practice satisfies that limit: it gives specific, operational meaning to the ambiguous qualifier “where feasible,” since strict FCFS continues to govern whenever the queue is uncongested and the practice operates only during peak contention. That is an exception at the margin the parties left undefined, not a substitute ordering rule.

The SLA’s merger clause does not change the analysis. A general integration clause shows at most that the writing is a complete and exclusive statement of its *terms*; it does not reach course-of-performance evidence, which the Code admits to explain and supplement a writing unless the parties have specifically negated that source.¹⁶⁴ Waiver supplies an alternative ground: even if the practice could not be reconciled with the term, ninety trading days without objection from FinCo, on which Orchestrion relied, could waive strict FCFS ordering.¹⁶⁵

The weighted factors support recognition:

- *Foreseeability* is high: fair-share arbitration under contention is a documented default of

¹⁶²See *supra* Part III.B. The nearest mandatory-law candidate is the Code’s non-waivable duty of good faith in performance, see *supra* note 88, but the practice raises no such concern: it allocates capacity evenhandedly among all Gold-tier tenants.

¹⁶³See U.C.C. § 1-303(e) (AM. L. INST. & UNIF. L. COMM’N 2022); *Nanakuli*, 664 F.2d at 780 (holding that the usage “formed an exception to, rather than a total negation of,” the express term). Nor is a threshold finding of ambiguity required. See U.C.C. § 2-202 cmt. 1(c) (AM. L. INST. & UNIF. L. COMM’N 2022) (rejecting any “requirement that a condition precedent to the admissibility” of practice evidence “is an original determination by the court that the language used is ambiguous”); *Columbia Nitrogen*, 451 F.2d at 9 (rejecting facial completeness as the test of admissibility); *Nanakuli*, 664 F.2d at 796 n.35 (“ambiguity is not necessary to admit usage evidence”).

¹⁶⁴See U.C.C. § 2-202(a) & cmt. 2 (AM. L. INST. & UNIF. L. COMM’N 2022); *Columbia Nitrogen*, 451 F.2d at 10–11 (addressing course of dealing and usage of trade; the extension to course of performance follows from § 2-202(a)’s grouping and cmt. 2); *Nanakuli*, 664 F.2d at 780; see also *supra* Part III.C.

¹⁶⁵See U.C.C. § 1-303(f) (AM. L. INST. & UNIF. L. COMM’N 2022) (providing that, “[s]ubject to Section 2-209, a course of performance is relevant to show a waiver or modification of any term inconsistent with the course of performance”); *id.* § 2-209(4) (“Although an attempt at modification or rescission does not satisfy the requirements of subsection (2) or (3) it can operate as a waiver.”); *Wis. Knife Works*, 781 F.2d 1280, 1286–87 (holding that conduct may operate as a waiver under § 2-209(4) only if the other party relied on it).

widely used schedulers, so a sophisticated tenant contracting for congestion-period compute had reason to anticipate it. The pattern was in any event visible to FinCo in its own tenant-level telemetry and API performance dashboards well before the disputed morning.

- *Regularity* is high within the window: queue logs show stable alternation during every peak-load episode, and FinCo’s systems began timing micro-batches to the scheduler’s cadence.
- *Attribution* points to both parties: to Orchestrio for deploying and maintaining the scheduler, and to FinCo for its agents’ learned adaptation, which solidifies the pattern as a mutual course of performance.¹⁶⁶
- *Materiality* is clear: the practice determined the allocation of compute and the timing of every Gold-tier job during congestion, including the delay to FinCo’s risk-control job. An event study bracketing the practice’s activation measures those effects tier-wide, showing reduced tail latency and the elimination of tenant starvation.
- *Verifiability* is robust: hash-anchored queue logs and versioned configuration files would let an independent expert replay the window using sealed scheduler binaries and FinCo’s job-submission patterns to confirm the practice and the adaptation to it.

The court grants the fair-share norm full recognition, using it not to rewrite the contract but to *interpret* “where feasible.” The months-long, undisputed pattern establishes a course of performance defining the parties’ obligations under congestion; with the term so construed, Orchestrio did not breach the SLA.¹⁶⁷

The outcome hinges on the contract. It would flip had the SLA instead read: “Within the Gold tier, scheduling shall be strict FCFS at all times, with no rotation, time-slicing, or fair-share

¹⁶⁶ See U.C.C. § 1-303(a)(2) (AM. L. INST. & UNIF. L. COMM’N 2022) (defining course of performance by the counterparty’s acceptance of or acquiescence in repeated performance “with knowledge of the nature of the performance and opportunity for objection to it”).

¹⁶⁷ See U.C.C. § 1-303(d) (AM. L. INST. & UNIF. L. COMM’N 2022) (making course of performance “relevant in ascertaining the meaning of the parties’ agreement”); RESTATEMENT (SECOND) OF CONTRACTS § 202(4) (AM. L. INST. 1981) (giving “great weight” to a course of performance “accepted or acquiesced in without objection”); *Nanakuli*, 664 F.2d at 794–96 (explaining that the parties’ actual performance of the contract—“the most important evidence of the agreement”—governed its meaning). See also *supra* Part III.C.

allocation under any circumstances. No contrary course of performance or usage of trade shall modify this term.” Such language removes any ambiguity and supplies the specific negation that the official comment to U.C.C. § 2-202 contemplates and that a general merger clause does not.¹⁶⁸ The practice would now directly contradict that term, and the Consistency gate would bar its recognition for interpretive purposes. Waiver—the escape § 2-209(4) preserves even against such clauses—would face a steep hurdle. Reliance on a practice that the parties expressly agreed could never modify the term is hard to call “reasonably induced.”¹⁶⁹ Orchestrio would likely be in breach.

Between these poles sits the case most likely to recur: an SLA that says only “first-come, first-served,” with a merger clause. As in the base case, the merger clause would not bar the evidence. But the practice would then confront a term with no undefined margin—supplanting FCFS exactly when the term has bite—and a court would likely hold that it negates rather than explains the term. The gate would still close, on the consistency of the practice rather than the sufficiency of the disclaimer.¹⁷⁰

Much in this case study is conventional: a bilateral agreement, a ninety-trading-day record, and telemetry that both parties’ systems could read. Two of the three frictions are absent—the practice was neither transient nor opaque. But the third, nonhuman origin, is what the case turns on. A course of performance requires that the accommodated party accept or acquiesce in the performance with knowledge of its nature and an opportunity to object; here it was FinCo’s systems, not its officers, that acquiesced. Whether an algorithm’s learned adaptation counts as its principal’s knowing acquiescence is the attribution question that Part II.D shows existing doctrine answers uneasily. The Attribution factor resolves it, supplying the finding that turns machine adaptation into party conduct.

¹⁶⁸ See U.C.C. § 2-202 cmt. 2 (AM. L. INST. & UNIF. L. COMM’N 2022) (explaining that, “[u]nless carefully negated,” prior dealings and usages “have become an element of the meaning of the words used”); *Columbia Nitrogen*, 451 F.2d at 10–11 (holding that a general merger clause was insufficiently specific to bar usage evidence).

¹⁶⁹ See *Wis. Knife Works*, 781 F.2d at 1287 (grounding § 2-209(4) waiver in reliance “reasonably induced and reasonable in extent”).

¹⁷⁰ See *Columbia Nitrogen*, 451 F.2d at 10–11.

C. Case Study 3: An Ambiguous Marketplace Practice

The third case study applies the doctrine where the outcome is not a simple pass or fail. In a high-speed digital advertising exchange run by a major platform, the AI-powered bidding agents of several large advertisers converge on a “deadline-edge bidding” practice.¹⁷¹ They all learn to submit their final, decisive bids just milliseconds before the exchange’s response timeout (‘tmax’). This practice, however, excludes smaller players whose off-the-shelf bidding agents cannot compete. The exclusion is architectural: large bidders gain an advantage through co-location and the engineering resources to minimize latency. Their agents can therefore use the full ‘tmax’ window for complex valuation before bidding at the last moment.

“AdCo” is a small company that, lacking these advantages, must bid early with less information or risk its bid arriving too late. AdCo sues the platform under state unfair competition law.¹⁷² It claims that the practice is a functional norm and that the platform, by designing a system where such a norm could emerge, has a duty to mitigate its exclusionary effects.

The interaction window is the three-month period preceding AdCo’s complaint—the period over which AdCo alleges the pattern stabilized and over which the platform’s logs are retained. The question is whether the claimed norm should be recognized and, if so, what legal consequences

¹⁷¹The “deadline-edge” bidding strategy is a rational response to exchange protocols that specify a maximum response time (‘tmax’) in real-time bidding (RTB) for digital advertising. See IAB TECH LAB, OPENRTB 2.6 § 3.2.1 (Apr. 2022), https://iabtechlab.com/wp-content/uploads/2022/04/OpenRTB-2-6_FINAL.pdf [<https://perma.cc/56AS-QHSL>] (defining ‘tmax’ in the BidRequest object as “Maximum time in milliseconds the exchange allows for bids to be received including Internet latency to avoid timeout”). Platforms emphasize latency reduction as a best practice. See GOOGLE FOR DEVELOPERS, *Latency Restrictions and Peering* (last visited Feb. 4, 2026), <https://developers.google.com/authorized-buyers/rtb/get-started/peer-guide> [<https://perma.cc/WG82-ZQ64>] (recommending network peering to minimize latency and bidding early enough to leave a buffer for network variance—guidance the emergent practice ran against; 85% of responses must arrive within ‘tmax’ to avoid throttling).

¹⁷²For concreteness, we assume a California-style unfair competition statute. See Cal. Bus. & Prof. Code § 17200 (West, Westlaw through ch. 101 of 2026 Reg. Sess.) (defining unfair competition to include “any unlawful, unfair or fraudulent business act or practice”); *id.* § 17204 (vesting public enforcement in the Attorney General, district attorneys, and certain city and county attorneys, while requiring of private plaintiffs injury in fact and lost money or property). For a claim like AdCo’s, grounded in competitive injury, the California benchmark tethers “unfair” to competition policy: it reaches conduct that “threatens an incipient violation of an antitrust law, or violates the policy or spirit of one of those laws . . . or otherwise significantly threatens or harms competition.” *Cel-Tech Commc’ns, Inc. v. L.A. Cellular Tel. Co.*, 973 P.2d 527, 544 (Cal. 1999). *Cel-Tech* confined the test to suits over a direct competitor’s practices, *id.* at 544 n.12—a posture the self-preferencing variant squarely presents.

follow.¹⁷³

The practice passes the Consistency gate, though not because the platform’s rules ignore bid timing. Those rules do address timing—imposing a maximum response time and throttling bidders who exceed it too often. But they set an outer bound; they do not prescribe when, within that bound, a bid must arrive. Deadline-edge bidding complies with the express timing term rather than contradicting it, and the gate closes only on contradiction.

The Legality Screen presents the harder question. On this record the practice clears the gate: it is not per se unlawful horizontal coordination, because nothing suggests an agreement among the bidders—each agent independently learned the same timing strategy from the same public protocol constraint. Neither a violation of mandatory public law nor the facilitation of one has been established.¹⁷⁴ But the practice sits close to the line. If the platform holds monopoly power in the exchange market, architecting an auction that creates de facto barriers to entry could constitute exclusionary conduct under Section 2 of the Sherman Act.¹⁷⁵ And if the platform marketed its auction as placing all bidders on an “equal footing” while knowing that its architecture produced the exclusion, the conduct could attract Federal Trade Commission (FTC) enforcement as an unfair or deceptive practice under Section 5 of the FTC Act.¹⁷⁶ Because the gate turns on established violations alone, these unresolved concerns cannot themselves defeat recognition. They bear instead on calibration: as Part III.C explains, public-policy concerns short of illegality are a reason to withhold the doctrine’s broader effects even where both gates are cleared.

The weighted factors are mixed:

- *Foreseeability* is met, but its incidence is asymmetric in a way the factor as stated does not capture. The ‘tmax’ parameter and the throttling threshold that together make deadline-

¹⁷³This posture is itself novel. In the preceding case studies, the proponent was a party to the practice it invoked. Here, the proponent is a party the practice excludes—though AdCo still bears the burden.

¹⁷⁴See *Gibson v. Cendyn Grp., LLC*, 148 F.4th 1069, 1083–84 & n.8 (9th Cir. 2025) (affirming dismissal of § 1 claims where plaintiffs alleged parallel algorithmic conduct without plausible allegations of an agreement or the pooling of nonpublic data), *cert. denied*, No. 25-1109, 2026 WL 1052046 (U.S. Apr. 20, 2026); *see also supra* note 121.

¹⁷⁵See 15 U.S.C. § 2 (2024).

¹⁷⁶See 15 U.S.C. § 45(a)(1)–(2) (2024) (declaring unfair or deceptive practices unlawful and empowering the Commission to prevent them).

edge bidding rational are published in the protocol specification that the exchange adopts and in the platform's own developer guidance, which documents the latency mechanics while recommending against bidding at the edge. A reasonable participant could therefore have anticipated the practice—the showing that AdCo, as proponent, must in any event make. What differs is not notice but the capacity to act on it: only the large bidders had the engineering resources to bid at the edge; AdCo could read the same documentation but still lack the means to do so. That asymmetry belongs to Attribution and to remedy.

- *Regularity* is satisfied within the interaction window. Timing histograms show stable clustering in the final milliseconds of the 'tmax' window during peak periods. The harder objection is the one Part III.B names—that the pattern is an artifact of an external constraint such as throttling rather than a genuine interactional practice. It fails on these facts. 'tmax' and the throttling threshold set the outer bound, but nothing in them dictates bidding at the window's edge. The clustering reflects what the bidders' agents learned to do within that bound.
- *Attribution* is contested, and must be assessed separately for each principal. As to the bidders, it is supported: their agents adopted and retained the strategy, and a principal who knowingly continues to deploy an agent and keep the resulting benefits may be charged with its behavior.¹⁷⁷ As to the platform, attribution does not run through the bidders' agents. The platform is answerable, if at all, for its own design choices—the tight 'tmax', the throttling rule, and the latency profile that made deadline-edge bidding the winning play. Yet those same parameters leave open whether the bidders adopted the practice or merely responded to one the auction's design imposed—a question AdCo's showing does not close.
- *Materiality* is clearly met: bid timing determines who wins, and counterfactual simulations isolate the effect of the final-millisecond submissions on auction outcomes.
- *Verifiability* is likewise met, though only because the records exist and have been produced. The proof lies wholly in the platform's auction logs, which no participant can reconstruct

¹⁷⁷ See *supra* Part III.B.

from its own data—an asymmetry that makes the discovery mechanisms discussed in Part V.A a precondition of satisfying the factor.

On balance, the practice functions as a norm, but the showing is modest. Given the Legality Screen’s concerns and the contested Attribution showing, the court grants narrow rather than full recognition, treating the practice’s existence as probative evidence in an unfair-competition analysis—an analysis for which the unresolved Section 2 concerns supply the tether *Cel-Tech* requires. Such recognition does not itself create a duty: it is evidentiary only. What the findings behind it supply, however, is a fact—that a stable, outcome-determinative practice emerged from the platform’s own auction parameters, and emerged predictably given them. Whether that fact makes the platform liable, and whether it owed any duty to mitigate, are questions for the underlying unfair-competition claim.¹⁷⁸ That is, the F-NORM test tells the court what happened in the auction; it does not tell the court who must answer for it.

The analysis would change character, though not simply in AdCo’s favor, if the evidence showed that the platform operated its own demand-side bidding agent inside the exchange—exposed to no network variance and no throttling risk, and so alone able to bid at the true edge every time. Such self-preferencing would reveal the practice as something other than organic emergence: an engineered outcome that advantages the platform’s own agent at competitors’ expense.¹⁷⁹ Where the platform holds the monopoly power hypothesized above, engineered self-preferencing of this kind is a paradigmatic form of exclusionary conduct under Section 2.¹⁸⁰

The Legality Screen’s open question would then resolve against the platform, and recognition

¹⁷⁸ Cf. *Verizon Commc’ns Inc. v. Law Offices of Curtis V. Trinko, LLP*, 540 U.S. 398, 411 (2004) (declining to expand “the few existing exceptions from the proposition that there is no duty to aid competitors”). The unfairness inquiry proceeds under the state statute’s own standard.

¹⁷⁹ See *States’ Second Am. Compl.* ¶¶ 236–45, 408, *Texas v. Google LLC*, No. 4:20-cv-00957 (E.D. Tex. Aug. 4, 2021), ECF No. 138; *States’ Third Am. Compl.* ¶¶ 297–98, 303–06, 376–77, *In re Google Digit. Advert. Antitrust Litig.*, No. 1:21-md-03010 (S.D.N.Y. Jan. 14, 2022), ECF No. 195 (alleging that Google used its dominant publisher ad server to give its own exchange and buying tools “Last Look” and similar information advantages over rival exchanges, and that its secret “Bernanke” program manipulated exchange auctions to the advantage of Google’s own buying tool).

¹⁸⁰ See *United States v. Grinnell Corp.*, 384 U.S. 563, 570–71 (1966) (stating the two elements of monopolization); *Spectrum Sports, Inc. v. McQuillan*, 506 U.S. 447, 456 (1993) (stating the elements of attempted monopolization); cf. *United States v. Google LLC*, 778 F. Supp. 3d 797, 873 (E.D. Va. 2025) (finding Google liable for monopolizing the publisher ad server and ad exchange markets and for unlawful tying).

would be categorically barred. AdCo would lose the evidentiary benefit it sought. But it would no longer need it: where conduct is independently unlawful, the antitrust claim itself does the work, and the doctrine steps aside. That is the framework working as designed.

D. Case Study 4: A Collusive Practice

The final case study illustrates the Legality Screen. Several competing online retailers all subscribe to the same third-party dynamic-pricing service. The service’s algorithm ingests each subscriber’s real-time, nonpublic inventory, pricing, and demand data, pools that data, and uses each firm’s confidential inputs to generate the prices it recommends to that firm’s competitors. The service promotes those recommendations as a way to avoid price wars and stabilize the market. In fact, the retailers almost always adopt the recommended prices; over several months, their prices for a popular product move in near-perfect lockstep, stabilizing at supracompetitive levels.¹⁸¹ A consumer class action alleges that the emergent price-matching behavior constitutes unlawful price-fixing under § 1 of the Sherman Act.¹⁸²

The retailers respond that the price alignment is not the product of an agreement but is instead a lawful, independently arrived-at market equilibrium.¹⁸³ Each firm, they contend, acted unilaterally in subscribing to the service and was free to accept or reject its recommendations. There was no meeting of the minds, no explicit cartel, and no coercion—just rational firms employing cutting-edge pricing technology. They ask the court to recognize the practice as a

¹⁸¹Pending antitrust litigation alleges that revenue-management software facilitates competitor coordination. Specifically, the plaintiffs in *In re RealPage, Inc., Rental Software Antitrust Litigation (No. II)*, No. 3:23-md-03071 (M.D. Tenn.), allege that landlords delegated pricing to a common algorithm that used nonpublic competitor data to stabilize rents at supracompetitive levels. See Statement of Interest of the United States, *supra* note 13, at 2. The MDL court largely sustained the § 1 claims at the pleading stage and has since preliminarily approved class settlements with numerous defendants. See *In re RealPage, Inc., Rental Software Antitrust Litig. (No. II)*, 709 F. Supp. 3d 478 (M.D. Tenn. 2023); Mem. Op. & Order at 1–2, *In re RealPage*, No. 3:23-md-03071 (M.D. Tenn. July 16, 2026) (reciting the preliminary approvals).

¹⁸²See 15 U.S.C. § 1 (2024) (declaring illegal “[e]very contract, combination in the form of trust or otherwise, or conspiracy, in restraint of trade”).

¹⁸³Where the agreement is explicit, algorithmic price-fixing has been charged criminally. See, e.g., Information at 3, *United States v. Topkins*, No. 3:15-cr-00201 (WHO) (N.D. Cal. Apr. 6, 2015), ECF No. 1 (charging that the conspirators “agreed to adopt specific pricing algorithms for the agreed-upon posters with the goal of coordinating changes to their respective prices”).

functional norm arising organically from autonomous agents reacting to competitive market signals, analogizing it to the benign coordination in Case Study 1. Recognition, on their theory, would supply affirmative evidence that the alignment was an equilibrium rather than the product of a “conscious commitment to a common scheme,” and so would help defeat the inference of agreement on which the plaintiffs’ § 1 claim depends.¹⁸⁴

The retailers have no agreement whose express terms the practice could contradict, and nothing in their vertical subscriptions purports to govern their prices. The Consistency gate asks nothing; analysis begins—and ends—with the Legality Screen.¹⁸⁵

The retailers’ conduct is found to have the function and effect of a horizontal price-fixing agreement, the paradigmatic per se violation of antitrust law.¹⁸⁶ The service operates as the hub of a hub-and-spoke arrangement: each retailer contracts vertically with the service, and the rim is inferred from their mutual, knowing contribution of nonpublic pricing data to a pool used to generate their competitors’ prices.

That rim is the agreement, inferred from the evidence—parallel conduct plus factors that tend to exclude independent action, such as the two-way exchange of confidential data and adherence to recommendations designed to “avoid price wars”—and once the combination is established, the per se rule forecloses inquiry into the intent behind the restraint.¹⁸⁷ While the retailers might

¹⁸⁴ See *Monsanto Co. v. Spray-Rite Serv. Corp.*, 465 U.S. 752, 764 (1984) (quoting *Edward J. Sweeney & Sons, Inc. v. Texaco, Inc.*, 637 F.2d 105, 111 (3d Cir. 1980)); *Matsushita Elec. Indus. Co. v. Zenith Radio Corp.*, 475 U.S. 574, 588 (1986) (requiring evidence “that tends to exclude the possibility” that the alleged conspirators acted independently).

¹⁸⁵ As is typical, the dispositive question is economic function, not technical form. See *Socony-Vacuum*, 310 U.S. 150, 223 (deeming the “machinery employed” for price-fixing “immaterial”); Statement of Interest of the United States, *supra* note 13, at 2 (“Automating an anticompetitive scheme does not make it less anticompetitive.”).

¹⁸⁶ See *Socony-Vacuum*, 310 U.S. at 223. For expository clarity, we assume a clear failure of the Legality Screen. The case law on “algorithmic collusion,” however, remains unsettled and highly sensitive to the pleadings. See cases cited *supra* notes 21, 120. Compare *Duffy v. Yardi Sys.*, 758 F. Supp. 3d 1283, 1291–98 (applying per se treatment to alleged horizontal price-fixing facilitated by revenue-management software), with *In re RealPage*, 709 F. Supp. 3d 478, 519–21 (declining per se treatment for novel ways of doing business and proceeding under the rule of reason). The Ninth and Third Circuits’ opposite outcomes turn on what was pleaded—both key on the pooling of nonpublic competitor data—not on divergent governing standards. See generally Amy Vegari, *Algorithmic Price-Fixing Cases Reflect Exacting Pleading Standard*, JD SUPRA (Aug. 25, 2025), <https://www.jdsupra.com/legalnews/algorithmic-price-fixing-cases-reflect-2043614/> [<https://perma.cc/H2DV-NP65>] (surveying the divergent outcomes).

¹⁸⁷ See *supra* note 121; see also *In re Text Messaging Antitrust Litig.*, 630 F.3d 622, 627–28 (7th Cir. 2010) (describing the “parallel plus” pleading standard on an earlier appeal in the same litigation). On the division over per se treatment for novel algorithmic mechanisms, see cases cited *supra* note 186. We take *Duffy*’s view: where the mechanism performs the function of horizontal price-fixing, *Socony-Vacuum*’s machinery rule controls. See *supra* note 119.

marshal evidence for the weighted factors—showing the price alignment was foreseeable from the service’s dashboards, highly regular in the price histories, attributable to each firm’s decision to adopt and maintain the common pricing service, material to their profits, and verifiable through logs—the weighted-factor inquiry is foreclosed at the threshold, though the underlying proof is not: whether the alignment reflects independent action remains an ordinary § 1 question. The court denies recognition—not because the pattern is not “functional” for the retailers, but because the function it serves is illegal.

The contrast with Case Study 1 is instructive. There, the practice solves a *coordination* problem—avoiding collisions—and enhances welfare; here, it solves the retailers’ *cooperation* problem, sustaining a supracompetitive price against each firm’s incentive to undercut, and harms consumers.¹⁸⁸ And where the drones converged on the altitude-for-direction practice from public broadcasts, the retailers converged on a common price from one another’s confidential data. Coordination that *enhances* welfare may warrant recognition; collusion that *suppresses* competition to extract rents must be condemned. This case demonstrates the outer limit of the doctrine: the framework will not lend interpretive force or evidentiary legitimacy to conduct the law forbids.

V. OPERATIONALIZING THE DOCTRINE: FROM COURTROOM TO CODE

Parts I–IV specified the F-NORM test, located the doctrine within private and public law, and applied it to four case studies. This Part turns from analysis to implementation. It shows that the doctrine is administrable largely through existing legal tools rather than novel institutional capacities.

Part V.A outlines evidentiary and procedural mechanisms for litigation, emphasizing rapid preservation, structured production, authentication, expert analysis, and a staged adjudication in which the burden tracks the character of the question. Part V.B turns to public governance, arguing

¹⁸⁸ See *supra* note 53.

that agencies enforcing existing statutes can reach practices no private litigant will challenge and that incentive design can generate the evidence the doctrine needs.

A. Evidentiary and Procedural Mechanisms

The F-NORM test is an evidence-driven inquiry. Its application depends on the ability of parties and courts to access, authenticate, and interpret the digital records of agent interactions.

The first procedural step in any dispute involving a claimed functional norm is therefore the preservation of the evidentiary substrate detailed in Part I.C. Given the ephemeral nature of system logs and telemetry data—where retention windows are set by operational policy and may run to days or weeks rather than years¹⁸⁹—parties must be prepared to issue immediate and specific litigation holds that cover not only communications but also interaction logs, configuration files, and model version histories. The risk of spoliation is acute and the sanctions potentially severe. Under Rule 37(e), a court may order measures to cure prejudice from lost ESI; harsher sanctions—including adverse-inference instructions or case-dispositive relief—require a finding that “the party acted with the intent to deprive another party of the information’s use in the litigation.”¹⁹⁰

Existing e-discovery rules are readily adaptable to AI agents. Under the Federal Rules of Civil Procedure, parties can and should request the production of data in structured, machine-readable formats (e.g., JSON or Parquet files with clear data dictionaries), rather than as static screenshots or PDFs.¹⁹¹ Proportionality principles can narrow requests to the specific interaction window

¹⁸⁹See NAT’L INST. OF STANDARDS & TECH., SPECIAL PUB. 800-92, *supra* note 79, at 5-9 (distinguishing “relatively short (days or weeks)” log-retention periods from “relatively long (months or years)” periods requiring archival planning).

¹⁹⁰FED. R. CIV. P. 37(e)(1)–(2). See FED. R. CIV. P. 37(e) advisory committee’s note to 2015 amendment (rejecting cases that had authorized adverse-inference instructions on a finding of negligence); *Zubulake IV*, 220 F.R.D. 212, 216–18 (establishing the duty to preserve electronic data once litigation is reasonably anticipated); *Zubulake V*, 229 F.R.D. 422, 431–34 (setting out counsel’s duty to issue, re-issue, and monitor compliance with a litigation hold); see also *supra* note 71 (introducing these preservation and production rules as applied to the evidentiary substrate).

¹⁹¹See FED. R. CIV. P. 34(b)(1)(C) (allowing a request to specify the form for ESI production); FED. R. CIV. P. 34(b)(2)(E)(ii) (requiring production in the form ordinarily maintained or a reasonably usable form if not specified); see also *Williams v. Sprint/United Mgmt. Co.*, 230 F.R.D. 640, 652, 655–56 (D. Kan. 2005) (ordering production of spreadsheets in their native format with metadata).

and the specific agents at issue, preventing burdensome “fishing expeditions” while ensuring that the data the F-NORM test requires is produced.¹⁹² Protective orders under Rule 26(c) safeguard sensitive information, and non-waiver orders under Federal Rule of Evidence 502(d) prevent its production from waiving privilege.¹⁹³

Given the technical complexity, expert witnesses will be central. Data scientists or computational social scientists can supply the qualified-person certification that makes electronic records self-authenticating under Rules 902(13)–(14), lay the foundation under Rule 901(b)(9) where reliability is contested, and establish an organization’s routine practice under Rule 406. They also serve as “translators” for the court, interpreting the evidentiary substrate using methods tied directly to the F-NORM factors: public-telemetry analysis for Foreseeability, time-series analysis for Regularity, version-history reconstruction for Attribution, event studies for Materiality, and deterministic replays for Verifiability.¹⁹⁴ This testimony is subject to the reliability standards of Rule 702 and *Daubert*.¹⁹⁵ In particularly complex or contentious cases, courts may also appoint a neutral expert under Rule 706 to assist the factfinder, or a special master under Rule 53 to manage the technical record and, where appropriate, recommend findings.¹⁹⁶

Courts can adopt a staged adjudication process. Where the record permits, a court can resolve the Consistency gate and the Legality Screen at summary judgment, reserving the more fact-

¹⁹² See FED. R. CIV. P. 26(b)(1).

¹⁹³ These tools are central to modern e-discovery. Rule 26(c) protective orders can limit who may view sensitive data and for what purpose, while a Rule 502(d) order allows parties to produce privileged material without effecting a waiver of privilege in the instant case or any other federal or state proceeding. See FED. R. CIV. P. 26(c); FED. R. EVID. 502(d); see also THE SEDONA CONFERENCE, *supra* note 71, at 150–51 cmt. 10.c (recommending that parties “protect against waiver by inadvertent production through agreements incorporated in a Fed. R. Evid. 502(d) order”).

¹⁹⁴ Records generated wholly by machine are ordinarily not hearsay; where human-supplied content is at issue, the business-records exception typically supplies the route to admission. See FED. R. EVID. 406, 803(6), 901(b)(9), 902(13)–(14), 1006; see also *supra* notes 66, 68, 74, 106.

¹⁹⁵ See FED. R. EVID. 702 (requiring, as amended in 2023, that the proponent of expert testimony demonstrate each admissibility condition to the court by a preponderance); *Daubert*, 509 U.S. 579, 592–95; *Kumho Tire*, 526 U.S. 137, 147–49 (extending the gatekeeping function to all expert testimony).

¹⁹⁶ See FED. R. EVID. 706; FED. R. CIV. P. 53; see also FED. JUD. CTR. & NAT’L RSCH. COUNCIL, REFERENCE MANUAL ON SCIENTIFIC EVIDENCE 6–8, 35, 329–30 (3d ed. 2011) (observing that “a court-appointed expert or a special master appointed pursuant to Rule 53 of the Federal Rules of Civil Procedure might be more desirable than a full-fledged *Daubert* battle among experts,” and surveying the court’s authority under Rule 706 and the practical considerations that govern appointment). The fourth edition (2025) retains no comparable treatment; the third remains the fullest discussion. See *infra* note 220.

intensive weighted factors for trial; where a gate turns on genuinely disputed predicate facts, the question goes to a factfinder instead.¹⁹⁷ Across both stages, the burden tracks the character of the question.¹⁹⁸ The gate questions are legal in character, even where their predicate facts are disputed: consistency with express terms is resolved by construction, and illegality may be raised by the opposing party or by the court on its own. No party bears a burden on a question of law, and where a gate instead turns on contested facts, the law that supplies the gate allocates the burden. In contrast, the proponent bears the burden of establishing the facts underlying the weighted factors by a preponderance of the evidence and of showing that, on balance, the practice functions as a norm.

B. Public Governance: Regulatory Enforcement and Incentive Design

A public-law backstop to private enforcement is necessary to ensure uniformity, protect third parties, and govern interactions where contracts are absent or adhesive.¹⁹⁹ While courts can develop the doctrine incrementally by applying common-law reasoning, regulatory agencies operating under existing statutes play a complementary role that courts cannot: policing harmful emergent practices no private litigant will challenge, and building the conditions that make the doctrine administrable.

Regulatory agencies such as the FTC can integrate the doctrine into their existing oversight and enforcement mandates. The Commission could, for example, issue an enforcement policy statement providing that a firm’s deployment of—or reliance on—an emergent practice that is itself unfair or deceptive to consumers may violate Section 5 of the FTC Act.²⁰⁰

¹⁹⁷ See FED. R. CIV. P. 56(a) (authorizing summary judgment on “the part of each claim or defense” where “there is no genuine dispute as to any material fact”); FED. R. CIV. P. 42(b) (providing for separate trials of separate issues); see also *supra* Part IV (analyzing each case study at the summary-judgment posture after discovery).

¹⁹⁸ See *supra* Part III.B.

¹⁹⁹ See *supra* Part II.E (establishing that mandatory public law sets the outer boundary of any practice-based doctrine); see also *supra* Part III.B (developing the Legality Screen as the doctrine’s public-law gate).

²⁰⁰ See 15 U.S.C. § 45(a)(1), (n) (2024) (prohibiting unfair or deceptive acts or practices and defining unfairness); Press Release, FED. TRADE COMM’N, *FTC Announces Crackdown on Deceptive AI Claims and Schemes* (Sept. 25, 2024), <https://www.ftc.gov/news-events/news/press-releases/2024/09/ftc-announces-crackdown-deceptive-ai-claims-schemes> [<https://perma.cc/W2K6-Y2HZ>] (announcing “Operation AI Comply,” an enforcement sweep against

Unfairness authority is particularly well suited to emergent conduct. Section 5(n) conditions liability on effects—substantial consumer injury that consumers cannot reasonably avoid and that is not outweighed by countervailing benefits—not on the respondent’s state of mind. The question that strains agreement-based theories therefore does not arise: the Commission need not find an agreement, only conduct and its consequences.²⁰¹ The F-NORM test supplies a framework for structuring a Section 5 investigation: the five weighted factors ask what the practice is, to whom it is attributable, and whether it can be proved, while the Legality Screen marks the public-law boundary.²⁰²

Enforcement need not be confined to consumer protection. Because Section 5 reaches unfair methods of competition as well as unfair or deceptive practices, the framework serves the Commission’s competition mission. The Justice Department’s Antitrust Division has already taken the position that algorithmic mediation is no shelter: prices fixed through the joint use of an algorithm are fixed prices all the same.²⁰³ The CFTC has begun asking the same questions.²⁰⁴ Here the Legality Screen does double duty. For a court, failure bars recognition. For an agency, the same finding identifies the practice as an enforcement target; state attorneys general enforcing unfair-competition statutes occupy that position.²⁰⁵

Beyond enforcement, public governance can also shape the incentives that determine whether the evidence the doctrine needs exists at all. Legislatures and regulators could, for instance, give an affirmative defense in enforcement actions to firms that voluntarily adopt and maintain rigorous

companies that “relied on artificial intelligence as a way to supercharge deceptive or unfair conduct”); *see also* FED. TRADE COMM’N, *Policy Statement Regarding the Scope of Unfair Methods of Competition Under Section 5 of the Federal Trade Commission Act* (Nov. 10, 2022), <https://www.ftc.gov/legal-library/browse/policy-statement-regarding-scope-unfair-methods-competition-under-section-5-federal-trade-commission> [<https://perma.cc/N26L-D26T>] (exemplifying the Commission’s use of policy statements to announce enforcement principles); *supra* note 124 and accompanying text (introducing the Commission’s unfairness authority as an outer boundary on practice-based recognition).

²⁰¹ *See supra* note 22 (identifying the intent-proof obstacle that an effects-based standard sidesteps).

²⁰² *See supra* Part IV.D (applying the Legality Screen to bar recognition of a collusive pricing practice while leaving benign convergence untouched).

²⁰³ *See supra* note 13.

²⁰⁴ *See supra* note 22 (citing the CFTC’s request for comment on artificial intelligence in the markets it regulates).

²⁰⁵ *See supra* note 172 (§ 17204’s allocation of public enforcement to the Attorney General and other public prosecutors).

logging, auditing, and transparency practices, keying the defense to recognized guidance such as NIST’s AI Risk Management Framework and log-management publications. The defense would place the burden of proving compliance on the firm invoking it—making the records themselves the price of the protection.²⁰⁶ This would create a powerful incentive for firms to generate the evidentiary substrate the doctrine depends on.²⁰⁷ Together, the litigation toolkit of Part V.A and the regulatory levers of Part V.B make the doctrine workable, largely without new institutions: courts already have the evidentiary rules, and agencies already have the statutes. The deeper effect is prospective: the doctrine’s evidentiary demands migrate upstream into system design—logging, audit trails, and the records the defense makes worth keeping—so that the courtroom’s requirements become code’s requirements. What remains is to answer the objections such a doctrine invites.

VI. ADDRESSING OBJECTIONS

This Part confronts three families of objections: *administrability*—indeterminacy, judicial capacity, and the evidentiary program’s burden and asymmetry; *legitimacy*—democratic authority, the incorporation critique, consent, and manipulation or capture; and *policy and timing*—innovation effects, prematurity, and jurisdictional variation. The organizing principle throughout is modesty: recognition of a functional norm is (i) interpretive in contract; (ii) probative, not dispositive, in tort; and (iii) subordinate to express terms and mandatory public law.

Critics will say the F-NORM test is too *indeterminate*: a weighted, multifactor standard with no

²⁰⁶This approach has analogues in recent state AI legislation, though the ground remains unsettled. See Tex. Bus. & Com. Code Ann. § 552.105(c), (e) (West 2026) (establishing a rebuttable presumption of reasonable care and barring liability in attorney general enforcement actions where the defendant discovers a violation through feedback, adversarial or red-team testing, state-agency guidelines, or an internal review process undertaken in substantial compliance with a recognized risk management framework); *id.* § 552.101(b) (providing no private right of action). Colorado enacted a comparable affirmative defense conditioned on curing a violation, but repealed it before it ever took effect, replacing it with a cure period rather than a defense. See Colo. Rev. Stat. § 6-1-1706(3), (6) (2024), *repealed and reenacted by* Act of May 14, 2026, ch. 131, sec. 1, 2026 Colo. Sess. Laws (S.B. 26-189). See also *supra* notes 79, 76 (the NIST log-management guidance and AI Risk Management Framework).

²⁰⁷See *supra* Part I.C (detailing the primary, secondary, and tertiary record classes that constitute the evidentiary substrate).

fixed weights invites unpredictable and inconsistent judicial outcomes, and unpredictability itself chills innovation.²⁰⁸ The concern is understandable but misplaced. The doctrine is intentionally a *standard, not a rule*, because the dynamism of AI environments requires flexibility; rules specified in advance would need to anticipate every configuration a system might adopt and would require amendment each time one changed. This is the classic rules-versus-standards tradeoff, and it cuts the doctrine's way: rules repay their costs where conduct is frequent and homogeneous, but standards dominate where conduct is heterogeneous—as algorithmic emergent practices are, varying across platforms and configurations and fast-changing within any one.²⁰⁹ Furthermore, the doctrine's discretion is disciplined by the test's evidence-based criteria—five weighted factors, each grounded in the concrete, digital records of the evidentiary substrate detailed in Part I.C, and two threshold gates anchored in the parties' express terms and in mandatory public law. The proponent bears the burden of establishing the facts underlying the weighted factors by a preponderance of the evidence and of showing that, on balance, the practice functions as a norm. And recognition is graduated, so that a marginal showing yields a narrow legal effect rather than an all-or-nothing liability event.²¹⁰ Accordingly, what the doctrine offers is not rule-like certainty—the price of the flexibility—but process predictability: parties know in advance what evidence will matter, what sequence the analysis follows, and who bears the burden.²¹¹ That kind of clarity matters: empirical work links policy uncertainty to delayed and depressed investment, and a bounded standard that settles expectations reduces that uncertainty.²¹²

Does the doctrine let machines make law? This objection mistakes the doctrine's modest scope. It does not grant AI agents any form of law-making authority. Rather, it does for algorithmic

²⁰⁸This issue finds an analogue in commercial law's incorporation debate. See *infra* note 214 and accompanying text.

²⁰⁹See Louis Kaplow, *Rules Versus Standards: An Economic Analysis*, 42 DUKE L.J. 557, 563–64, 616–17 (1992) (describing standards as suited to “more heterogeneous behavior, in which each relevant type of act may be rare,” and as easier to keep current as conditions change).

²¹⁰See *supra* Part III.B–C (specifying the test's sequential structure, the proponent's burden, and graduated recognition).

²¹¹See Kaplow, *supra* note 209, at 615–16 (locating predictability in common knowledge and practice even absent formally articulated rules).

²¹²See, e.g., Scott R. Baker, Nicholas Bloom & Steven J. Davis, *Measuring Economic Policy Uncertainty*, 131 Q.J. ECON. 1593, 1596, 1627 (2016) (documenting the association between policy uncertainty and delayed or reduced investment).

practice what the common law has long done for human custom: it recognizes a fact (a stable, functional pattern) and gives it calibrated legal effect (interpretive and evidentiary), without elevating it above bargains or statutes, for the limited purposes of interpreting private agreements, filling gaps, and assessing standards of care.²¹³ Even the warrant for that modest effect is party autonomy, not machine authority: the Attribution factor asks whether the practice traces to a legally responsible principal that chose to deploy it, such that recognition rests on the principal's conduct. And it does not displace legislatures: it yields wherever positive law speaks—where a legislature prefers a rule, its rule governs.

The continuity answer must, however, face the leading critique of the very tradition it invokes. Lisa Bernstein has argued that the Code's incorporation strategy rests on two mistaken premises: the unwritten norms transactors follow while cooperating are relationship-preserving accommodations, not the end-game terms they would want a tribunal to enforce, and judicial resort to unwritten usage generates interpretive uncertainty rather than predictability.²¹⁴ If she is right about human merchants, extending the incorporation strategy to machine practices might seem to compound the error. But the critique's two premises cut differently here. The uncertainty premise is about evidence: courts reconstructing a usage from partisan recollection and dueling experts will get it wrong often enough to destabilize expectations. The F-NORM test does not reconstruct practices from recollection; it admits only practices demonstrated in the systems' own records—logs, configuration histories, telemetry—with existence and scope proved, not surmised.²¹⁵ Where experts still contend, they contend over records the court can inspect, not recollections it cannot. The relational premise is about consent: cooperative forbearance should not harden into enforceable obligation. The doctrine internalizes rather than overrides that point: recognition, when granted, is interpretive and evidentiary—it colors the meaning of

²¹³Courts routinely recognize “usage of trade” and “course of performance” to interpret agreements, subject to a settled hierarchy in which express terms ordinarily prevail and mandatory public law always does. *See* U.C.C. § 1-303(d)–(f) (AM. L. INST. & UNIF. L. COMM'N 2022); RESTATEMENT (SECOND) OF CONTRACTS §§ 202(4), 222 (AM. L. INST. 1981).

²¹⁴*See* Bernstein, *supra* note 45, at 1768, 1802, 1805–06.

²¹⁵*See supra* Part I.C.

undertakings the parties chose and is never the immanent law Llewellyn hoped courts would enforce—and where the parties have addressed a practice in their written terms, the Consistency gate leaves them free to bar its recognition outright. And a single act of forbearance is not a practice: Regularity measures repetition within the window.²¹⁶ The Article’s incorporation claim is thus narrower than the tradition Bernstein criticizes: not that courts should go looking for the norms latent in commercial life, but that when a machine-speed practice is provable, attributable, and consistent with the parties’ own bargain, courts may read the bargain in its light.

A neighboring objection sounds in *consent*: if no human at either firm ever observed an emergent practice, how can it inform what the parties agreed to or should have expected? The premise—that legal consequence requires contemporaneous human awareness—was surrendered a generation ago. Electronic-transactions law binds parties to contracts formed by their electronic agents without any human review or awareness, and agency law has never required a principal to watch each act performed by its chosen instruments—attribution follows the decision to deploy, not the fact of observation.²¹⁷ The F-NORM test is more demanding than that baseline, not less. Foreseeability asks whether a reasonable party, in context, could have anticipated the practice’s emergence—through documented behaviors, announced parameters, or regularities observable in public telemetry—so the test charges a party with what it had the means to know, whether or not anyone looked.²¹⁸ Nor does recognition require opening the black box. That notice runs to behavior: the test turns on what a system did—captured in the auditable outputs and records courts already evaluate—not on why its model did it.²¹⁹ What consent requires, the test supplies: notice through discoverability, and responsibility through attribution.

Can generalist judges adjudicate disputes over complex algorithmic behavior? The doubt about *judicial capacity* mistakes how courts absorb technical domains: not by making judges data scientists, but by building procedures that route technical questions to those equipped to

²¹⁶ See *supra* note 84; see also *supra* Part III.B–C.

²¹⁷ See *supra* notes 107, 132 and accompanying text.

²¹⁸ See *supra* Part III.B (the Foreseeability factor and the evidence that establishes or rebuts it).

²¹⁹ See *supra* note 66 and accompanying text; see also *supra* Part I.C.

answer them—as claim construction did in patent law and expert economics did in antitrust. The doctrine channels technical issues through a mature procedural toolkit: the Federal Judicial Center and National Academies’ *Reference Manual on Scientific Evidence*, court-appointed experts under Rule 706, and special masters under Rule 53.²²⁰ Expert testimony is vetted through the familiar gatekeeping standards of Rule 702, *Daubert*, and *Kumho Tire*.²²¹ That gatekeeping cuts both ways—in many cases the first motion will be a Rule 702 challenge to the proponent’s own expert—but the exposure is discipline, not embarrassment: a practice that cannot be established through admissible, reliable analysis is a practice the test is built to reject. Nor does the ordinary case need the toolkit’s heavier instruments: appointments under Rules 706 and 53 remain exceptional, and the median dispute rests on routine admissibility—the evidentiary substrate of native logs, configuration histories, and telemetry is authenticated under Rules 901 and 902 and summarized under Rule 1006, as Part V details.²²² Technical complexity is therefore not a reason to ignore operative facts that the law can already admit and evaluate.

The evidentiary program draws a twofold objection. One half is *burden*: the doctrine’s logging and audit demands implicate privacy, strain data-minimization and retention-limit principles, and force disclosure of trade secrets and other sensitive operational data. The other is *asymmetry*: in platform markets one side controls the logs, telemetry, and configuration histories. This one-sided custody threatens to make the test unfair or unworkable.

Both concerns are manageable. Take burden first. Parties can employ tiered retention windows keyed to transactional risk, minimize fields to those needed for attribution and replay, and obtain protective orders under Rule 26(c) to safeguard sensitive materials through limited access, sealed appendices, and strict confidentiality protocols.²²³ As Part V explains, proportionality, interaction-window scoping, and structured data production (rather than unfiltered data dumps) reduce burden

²²⁰ See NAT’L ACADS. OF SCIS., ENG’G & MED. & FED. JUD. CTR., *REFERENCE MANUAL ON SCIENTIFIC EVIDENCE* 1409, 1481 (4th ed. 2025) (the new reference guides on computer science and artificial intelligence); see also *supra* note 196 (the Rule 706 and Rule 53 machinery).

²²¹ See *supra* note 195 (collecting FED. R. EVID. 702, *Daubert*, and *Kumho Tire* on the gatekeeping standard).

²²² See *supra* note 194 (collecting the admissibility authorities under FED. R. EVID. 803(6), 901(b)(9), 902(13)–(14), and 1006).

²²³ See FED. R. CIV. P. 26(c)(1)(A), (D)–(H).

while preserving probative value.²²⁴ Where applicable, data-minimization law reinforces rather than contradicts this approach: the principle that justifies limiting collection to what is reasonably necessary for a disclosed business purpose supplies the same logic for limiting forensic disclosure to what is necessary for adjudication—and the GDPR itself exempts processing necessary for legal claims from its erasure right.²²⁵

As is typical in platform litigation, asymmetry is the harder half: every litigation remedy operates only on records that already exist. For the records that do exist, the tools are at hand: the requesting party can specify the form of production and proceed by targeted sampling, which delivers the relevant subset without unbounded burdens;²²⁶ Rule 37(e) supplies graduated consequences for spoliation—curative measures on a finding of prejudice, adverse-inference instructions only on a finding of intent to deprive—once the duty to preserve attaches;²²⁷ and for particularly complex datasets or disputes over technical production, a Rule 53 special master can supervise discovery, conduct neutral replay tests, or resolve methodological disputes.²²⁸

What is distinctive here is not who holds the records but what they are: a functional norm exists only as a pattern of machine behavior, so the substrate does not merely evidence the practice—it very nearly constitutes it. That substrate is rarely one party’s: a norm comprises the conduct of many, and where it emerges from public signals each participant can log it. Where a platform sits between them, though, the asymmetry runs to creation rather than custody: the party that configures logging decides, before any dispute arises, what is recorded, at what granularity, and for how long. The test answers the resulting evidentiary risk with machinery that ordinary discovery law lacks. Verifiability assigns the risk of missing records to whoever asserts the practice—an

²²⁴ See *supra* notes 191–192 and accompanying text.

²²⁵ See Regulation (EU) 2016/679 of the European Parliament and of the Council, art. 5(1)(c), 2016 O.J. (L 119) 1, 35 (General Data Protection Regulation) (data minimization); *id.* art. 17(3)(e), at 44 (excepting from the erasure right processing necessary “for the establishment, exercise or defence of legal claims”); see also Cal. Civ. Code § 1798.100(c) (West 2025) (requiring that a business’s collection, use, retention, and sharing of personal information be reasonably necessary and proportionate to disclosed purposes).

²²⁶ See FED. JUD. CTR., MANUAL FOR COMPLEX LITIGATION (FOURTH) § 11.493 (2004) (statistical sampling in discovery).

²²⁷ See *supra* notes 71, 190.

²²⁸ See FED. JUD. CTR., MANUAL FOR COMPLEX LITIGATION (FOURTH), *supra* note 226, §§ 11.446, 11.51–11.52 (discovery of computerized data, court-appointed experts and technical advisors, and special masters).

unprovable norm fails no matter which side invokes it. One-sided custody confers no advantage in establishing a norm, and a custodian that keeps no records prices itself out of the doctrine's benefits, making the records worth keeping.²²⁹ What is familiar in the objection thus meets familiar tools; what is novel is priced into the test's own design.

Could sophisticated actors *manipulate* the doctrine itself—engineering artificial “norms” as pretexts for liability shields or anticompetitive safe harbors?²³⁰ The test meets a manufactured pattern in sequence, and the sequence is the safeguard. The Legality Screen comes first and is dispositive: as the price-fixing case study demonstrates, an emergent practice that amounts to illegal collusion fails at the threshold, regardless of how well it satisfies the other criteria.²³¹ A pattern that is lawful but manufactured then faces the factors, and the proponent's own burden. Evidence that it was generated by hidden configurations, nonpublic A/B testing, or parameters not reasonably discoverable rebuts Foreseeability; the absence of genuine counterparty reliance weakens both Regularity and Materiality; and a pattern with no operational function beyond creating a legal artifact fails Materiality outright—counterfactual analysis showing that the alleged norm did not meaningfully affect outcomes defeats recognition.²³² Even a scheme that survives all three earns only graduated recognition, calibrated to what the record shows. The worry also has a structural form: platform incumbents author the architectures in which practices emerge, and might thereby shape the norms the doctrine would recognize. But recognition confers no ownership. A practice is read against every participant alike, including its architect; express terms let any counterparty negate it; and where authorship shades into exclusion or collusion, the Legality Screen—along with the public enforcement Part V.B describes—remains in play. Manufactured or inherited, a pattern binds only insofar as it is provable, lawful, and consistent with the parties' own bargain.

Some may worry that the doctrine will *stifle innovation* by creating new legal risks that

²²⁹ See *supra* note 206 and accompanying text; see also *supra* Part III.B (the Verifiability factor).

²³⁰ Cf. Fletcher, *supra* note 22, at 263–64 (describing how algorithmic and self-learning trading strategies strain manipulation doctrine built on proof of human intent).

²³¹ See *supra* Part IV.D; *supra* note 14.

²³² See *supra* Part III.B.

freeze development by making companies afraid to update or improve their systems lest they disrupt a recognized norm and incur liability. The fear misconceives what recognition attaches to. A functional norm rests on a record of regular practice, and it counts against a deviating firm only while others still reasonably rely on it. A firm that wants a different practice changes course in a way that respects that reliance—announcing the change, accommodating those still following the norm during the transition, or withdrawing from the interaction altogether. Once the recognized practice ceases, the old norm lapses with its window, and any successor practice must earn recognition on its own record.²³³ If the altitude-for-direction norm is to end, the fleets can announce new separation rules or stop flying the corridor; after enough time, there is nothing left to recognize. Custom has always worked this way—trade usage need not be ancient, and the practices the law recognizes are expected to change over time.²³⁴ And updating away from a bad practice is not just permitted but sometimes required.²³⁵

Others may argue the doctrine is *premature*—that courts should wait for more cases to develop organically before codifying a recognition framework. This position undervalues the costs of a doctrinal vacuum. Without a coherent framework, early cases are resolved *ad hoc*, creating conflicting precedents across jurisdictions—already visible in the first wave of algorithmic-coordination cases, where courts divide even over the governing mode of analysis.²³⁶ Parties cannot structure their conduct against unpredictable judicial improvisation.²³⁷ The F-NORM test provides a common analytical vocabulary—courts retain discretion in applying the factors, but they begin from shared ground rather than constructing an analysis anew in each case. Moreover, the doctrine is designed to be adopted incrementally: courts can apply it case-by-case, and parties can contract into or out of it.²³⁸ The framework enables, rather than forecloses, organic development.

Finally, critics may worry that the doctrine will be applied inconsistently across states and

²³³ See *supra* Part III.B (defining Regularity over the relevant interaction window).

²³⁴ See RESTATEMENT (SECOND) OF CONTRACTS § 222(1) & cmt. b (AM. L. INST. 1981); see also *supra* note 25.

²³⁵ See *supra* note 145.

²³⁶ See *supra* notes 21, 186 (collecting divergent outcomes in the revenue-management-software cases).

²³⁷ See *supra* note 16 (the *Peraire-Bueno* jury's deadlock over which rules governed).

²³⁸ See *supra* Part III.C; *supra* Part V.A.

nations—*jurisdictional variation* that invites forum shopping and undermines predictability. Several considerations, however, allay these concerns. First, in contract, parties can solve the problem themselves, with tools they already deploy: choice-of-law and forum-selection clauses. Recognition is interpretive—an incident of the agreement, not a freestanding regime—so it travels with whatever law those clauses name. Second, in tort, where parties cannot choose the governing law, recognition is only evidence of the standard of care under the forum’s own custom doctrines—it adds nothing to the ordinary state-to-state variation of negligence law.²³⁹ Third, the doctrine works from materials that are widely shared. Within the United States, UETA, adopted in nearly every jurisdiction, and the federal E-SIGN Act, which supplies a national baseline, both recognize that electronic agents can form binding contracts;²⁴⁰ agency law supplies the attribution framework;²⁴¹ and contract and tort law’s practice-recognition doctrines—usage of trade, course of performance, custom as evidence—are foundational features of American law.²⁴²

For cross-border transactions, the U.N. Convention on Contracts for the International Sale of Goods (CISG) supplies the structural template: absent contrary agreement, it treats parties as impliedly bound by usages of which they knew or should have known and which are widely known and regularly observed in the relevant trade. That is a notice inquiry paired with an objective regularity requirement—the pairing the F-NORM test repeats in Foreseeability and Regularity. And what the Convention presupposes—humanly legible practices—the doctrine’s evidentiary substrate delivers for machine-speed ones.²⁴³ What remains, then, is ordinary variation of the kind the law already manages: the doctrine builds on widely shared foundations and creates no new conflicts problem of its own.

Taken together, the objections sharpen the doctrine rather than defeat it. Every answer draws on the same commitments: evidence before effect, so that nothing takes effect that cannot be

²³⁹ See *supra* Part III.C.

²⁴⁰ See authorities cited *supra* note 107; see also *supra* note 108 (noting the New York holdout).

²⁴¹ See authorities cited *supra* note 132.

²⁴² See *supra* Part II.

²⁴³ See CISG, *supra* note 96, art. 9(2); see also *supra* Part II.A (the Convention’s legibility presupposition).

proved from records; hierarchy before recognition, so that no practice is recognized against a contrary express term or mandatory public law; and calibration before liability, so that recognition does interpretive and probative work scaled to the strength of the showing. A doctrine so bounded claims modest ground: a common-law method for giving effect to facts that already govern conduct, available to courts now and open to refinement by legislatures later.

CONCLUSION

Where no pre-ordained rule exists, AI agents improvise de facto rules—who yields, how to pace, how to ration scarce slots—that are invisible to a legal system built for human custom and written contracts. And the law cannot incorporate what it cannot see.

The doctrine of functional norms provides the necessary lens. Its F-NORM test supplies a principled, evidence-based framework: a practice should be recognized when it clears both gates and, on balance, is foreseeable, regular, attributable, material, and verifiable—fully or narrowly, as the strength of the showing warrants—and should not when it contradicts express terms, violates mandatory public law, or fails to function as a norm.

In contract, recognized norms can serve interpretive and gap-filling functions: they can give particular meaning to the parties’ terms and supplement silent agreements, subject to the Code’s hierarchy of sources. A course of performance can show a waiver or modification of an inconsistent term.²⁴⁴ In tort, recognized norms can serve as probative but not dispositive evidence of the standard of care. As with human custom, courts retain the authority to find that a norm itself was negligent or that an entire calling lagged behind reasonable prudence.²⁴⁵ In procedure, the doctrine channels proof through established evidentiary mechanisms—ESI production in a specified or reasonably usable form, authentication under Rules 901 and 902, business records and summaries under Rules 803(6) and 1006, and expert testimony subject to *Daubert* and Rule 702

²⁴⁴ See U.C.C. §§ 1-303(d)–(f), 2-209(4) (AM. L. INST. & UNIF. L. COMM’N 2022); see also *supra* notes 85 (the hierarchy of sources), 87 (waiver by course of performance notwithstanding a no-oral-modification clause).

²⁴⁵ See *The T.J. Hooper*, 60 F.2d 737, 740.

gatekeeping.²⁴⁶

The doctrine remains subordinate to existing law. It does not override express terms: the Consistency gate bars recognition of a practice that directly contradicts a clear, express term of a governing agreement. Private ordering retains primacy, and parties can disclaim reliance on emergent practices through specific contractual language that carefully negates them, as the Code’s official commentary requires.²⁴⁷ It also does not sanitize unlawful conduct: the Legality Screen bars recognition of practices that violate antitrust law, civil rights statutes, consumer protection rules, or other mandatory public law. An emergent practice that amounts to, or facilitates, such a violation fails at the threshold, regardless of its regularity or verifiability.²⁴⁸ It confers no legal personhood on AI²⁴⁹ and no immunity from liability: recognition is evidentiary, not exculpatory.

Operationally, the framework is administrable. Courts can order targeted, native-format productions for the relevant interaction window, resolve the two gates at summary judgment where the record permits, and instruct juries that a recognized norm is evidence of care, not its measure.²⁵⁰ Parties can adopt, limit, or condition practices through express terms drafted against the background rule the doctrine supplies. Regulators can target harmful equilibria under existing unfair-practices authority, using the F-NORM test to separate benign coordination from harmful market-wide practices, and can incentivize logging and transparency to make accountability possible.²⁵¹

By allowing the law to see the operative practices of AI agents without elevating those practices above express contracts or mandatory public law, the doctrine equips courts to engage with the

²⁴⁶ See *supra* Part V.A; see also *supra* notes 194 (the hearsay and authentication treatment of machine-generated records), 195 (the amended Rule 702 gatekeeping standard).

²⁴⁷ See U.C.C. § 2-202 cmt. 2 (AM. L. INST. & UNIF. L. COMM’N 2022); *Columbia Nitrogen*, 451 F.2d 3, 10–11.

²⁴⁸ See *supra* notes 119, 123, 124; see also *supra* Part IV.D (price-fixing case study).

²⁴⁹ See, e.g., Joanna J. Bryson, *Robots Should Be Slaves*, in CLOSE ENGAGEMENTS WITH ARTIFICIAL COMPANIONS: KEY SOCIAL, PSYCHOLOGICAL, ETHICAL AND DESIGN ISSUES 63, 63 (Yorick Wilks ed., 2010) (arguing that robots “should not be described as persons, nor given legal nor moral responsibility for their actions”); Katherine B. Forrest, *The Ethics and Challenges of Legal Personhood for AI*, 133 YALE L.J.F. 1175, 1175–76 (2024) (discussing the question the extension of legal personhood to AI will pose for judges).

²⁵⁰ See *supra* Parts III.C and V.A; *The T.J. Hooper*, 60 F.2d at 740.

²⁵¹ See *supra* Part V.B; see also *supra* notes 124 (unfair-practices authority), 206 (state-law analogues keying safe harbors to recognized risk management frameworks).

de facto rules that govern AI agent interactions while holding innovation and accountability in balance. The alternative is a choice between two flawed extremes: *under-recognition*, which treats emergent practices as legal nullities, undermining liability and contractual certainty in algorithmic markets; and *over-recognition*, which—because a successful practice is not thereby fair, efficient, or lawful—risks laundering harmful or collusive conduct as legitimate “custom.” The doctrine avoids both.